

CHAPTER 4

Chapter Goals:

- To describe numerical and graphical techniques for exploring a model's behavior

Chapter Concepts:

- Initial conditions
- Variable over time plots
- General solution
- Variable versus variable plots
- Phase-plane diagrams
- Vector-field plots
- Null clines

Numerical and Graphical Techniques— Developing a Feeling for Your Model

4.1 Introduction

In the next few chapters we explore various mathematical techniques commonly used to analyze models. Before launching into these techniques, it is worthwhile having a sense of what to expect from a model. This chapter describes how to get a feel for a model's behavior through graphs. The basic approach is to specify numerical values for all of the parameters, and for the initial values of each variable, and then to use the model's equations to predict what happens over time.

If, for your organism(s), you knew the exact values of each parameter and the initial values of each variable, then a graph might be all that you need to answer your questions. Most of the time, however, biologists don't have such precise knowledge. Furthermore, we often want to use a model to understand general processes occurring in nature rather than a specific process in a specific species. The downside of the numerical techniques discussed in this chapter is that they are not well suited for understanding the behavior of a model in general terms. For example, they cannot answer questions like: Do the variables always grow under all conditions? When do the variables tend toward one value versus another? What are the conditions under which the variables change over time without ever settling down?

Even when your goal is to answer general questions, however, exploring numerical examples is worthwhile. Comparing mathematical results to numerical examples is a great way to check your calculations. If you find that a variable goes to zero in a graph under a particular set of conditions but your mathematical analysis suggests that this cannot happen, then clearly an error has crept into either the code generating the graph or the mathematical analysis. Graphs illustrating the dynamics of the model in particular cases can also guide which mathematical analyses to perform. Humans are great at seeing patterns, and it is often easier to prove that a certain pattern exists than to analyze equations blindly without any sense of how they might behave.

Although some of the graphs we discuss can be drawn by hand, computers are tremendously helpful in the numerical analysis of models. In particular, software packages like *Mathematica* (Wolfram Research), Maple (Waterloo Maple Inc.), and Matlab (MathWorks, Inc.) allow us to enter equations and generate graphs in a matter of a few lines of computer code. These “mathematical software packages” are essentially fancy calculators that manipulate symbols as well as numbers. In the on-line material, we provide introductory

notebooks for using mathematical software packages (“Labs”). If you have access to the relevant software, you should familiarize yourself with some of the more commonly used commands by working through the first Lab. We also provide the code used to generate each type of graph in the on-line supplementary material for Chapter 4.

For dynamical equations, there are three main types of graphs. In section 4.2, we discuss graphs that plot a variable as a function of time (i.e., as a function of the independent variable). Such graphs illustrate the dynamics of a variable for a given set of parameters. In section 4.3, we discuss graphs that plot the change in a variable as a function of the variable itself (i.e., as a function of the dependent variable). Such graphs clarify the conditions under which a variable grows or shrinks. In section 4.4, we discuss graphs that plot one variable as a function of another variable (i.e., as a function of another dependent variable). Such graphs help us see how interactions among the variables affect their dynamics.

4.2 Plots of Variables Over Time

Variables in discrete-time models are straightforward to graph as functions of time. Because they can be written as recursion equations (e.g., giving $n(t + 1)$ as a function of $n(t)$), discrete-time models can always be “iterated.” Given the starting values for each variable at time 0 (the *initial conditions*) and values for each of the parameters, the recursion equations determine the values of the variables at time 1. We can then plug these new values into the recursions again to find the values of the variables at time 2, etc. Such iteration of a discrete-time model can be used to obtain numerical values for the variables at any future point in time, given specific parameters and initial conditions. Having done so, a variable of interest can be plotted on the vertical axis against time on the horizontal axis.

The *initial conditions* (or *starting conditions*) are the numerical values of every variable at the initial time point.

4.2.1 The Dynamics of the Exponential Population Growth Model

Consider the recursion equation for exponential growth, $n(t + 1) = R n(t)$, and the initial condition $n(0) = 1000$ (i.e., a population of size 1000 at time $t = 0$). If the number of surviving offspring per parent, R , is 1.01, the recursion tells us that $n(1)$ will be $1.01 \times 1000 = 1010$. With $n(1)$ in hand, the recursion then tells us that $n(2) = 1.01 \times 1010 = 1020.1$. Continuing in this fashion, we get the series of population sizes graphed in Figure 4.1 (bottom curve in panel a). We can repeat this process using different values of the parameter, R , to see how this parameter affects the shape of the graph (Figure 4.1). A key point emerges from the figure: exponential growth causes dramatic increases in population size for large values of R , but the population size declines for those values of R that are less than one. This makes intuitive sense: if each individual leaves fewer than one surviving individual in the next time step, then the population should decrease over time until it goes extinct.

Variable over time plots illustrate the behavior of the variable of interest (vertical axis) versus time (horizontal axis).

The recursion equation for exponential growth is simple enough that we can even use the above iterative procedure without specifying parameter values.

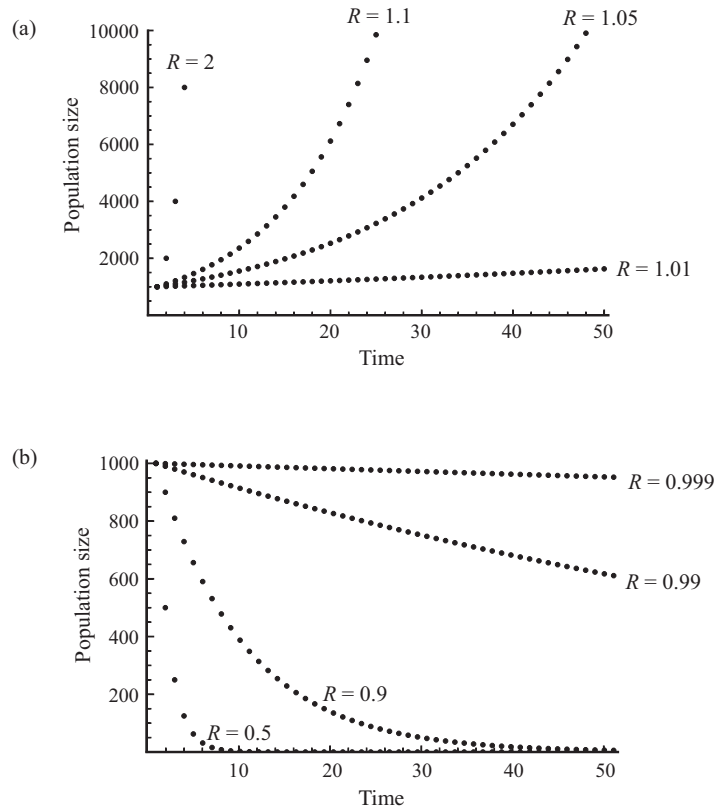


Figure 4.1: The dynamics of the exponential model in discrete time. The recursion equation (3.1b) is iterated for 50 time steps, giving the population size, $n(t)$, on the vertical axis as a function of time on the horizontal axis for various values of R (the number of surviving individuals at $t + 1$ per individual at t). The initial population size is $n(0) = 1000$.

Starting with a population of size $n(0)$ at time 0, there will be $R n(0)$ individuals after one time unit. After the second time unit, $n(2) = R n(1)$, but we already know that $n(1) = R n(0)$, and therefore $n(2) = R^2 n(0)$. Repeating this procedure, after three time units there will be $R^3 n(0)$ individuals. Every time we iterate this recursion, a 1 is added to the exponent of R . Therefore, after t time units, the number of individuals is

$$n(t) = R^t n(0). \quad (4.1)$$

A *general solution* to a dynamic model is an equation giving the value of a variable at any future point in time, as a function of the parameters, the initial conditions, and the amount of time that has elapsed.

This is the *general solution* to the exponential growth model, and it gives the population size at any future point in time for any value of the parameter R without having to calculate intermediate population sizes along the way.

The general solution allows us to determine exactly when the population will grow or shrink without having to explore all possible values of R numerically. $R^t n(0)$ always equals $n(0)$ when R equals one, it declines as time passes whenever R is less than one, and it increases as time passes whenever R is greater than one.

Does modeling population growth in continuous-time generate different predictions? To address this question graphically requires that we predict future population sizes using the differential equation $dn/dt = r n(t)$. That is, we must “solve” the differential equation so that we know $n(t)$ itself at time t , and not just the rate at which $n(t)$ changes over time. Just as with discrete-time recursion

equations, differential equations can be solved analytically or numerically. We will describe analytical methods for solving differential equations in Chapter 6. For now, we solve the exponential growth model in continuous time using an educated guess. While this sounds like a crazy idea, many differential equations have been solved in this way.

In the continuous-time model for exponential growth, we might guess that the solution is similar to the discrete-time solution (4.1): $n(t) = R^t n(0)$. If we take the derivative of this function for $n(t)$ with respect to t (using Rule A2.7 in Appendix 2), we get $dn/dt = \ln(R) R^t n(0)$, which is equivalent to $dn/dt = \ln(R) n(t)$ according to our purported solution, $n(t) = R^t n(0)$. This differential equation is not exactly the same as the one that we are trying to solve, $dn/dt = r n(t)$. But if we set $\ln(R)$ to r (equivalently, R to e^r), these two differential equations match up. Therefore, we can correct our initial guess by replacing R with e^r :

$$n(t) = e^{rt} n(0). \quad (4.2)$$

If we take the derivative of (4.2) with respect to time, we get $dn/dt = r e^{rt} n(0)$. We can then substitute in solution (4.2) to regain $dn/dt = r n(t)$, confirming that (4.2) is indeed the general solution. Equation (4.2) clarifies why this is called the “exponential growth” model; the population size changes exponentially with time. Equation (4.2) can then be plotted over time, resulting in curves that pass through the dots shown in Figure 4.1 (as long as we replace R with e^r ; see Lab exercise on plotting functions).

4.2.2 The Dynamics of the Logistic Population Growth Model

For most models, it is not so simple to obtain a general solution (see Chapter 6). For example, if you try to iterate the logistic equation (3.5a),

$$n(t + 1) = n(t) + rn(t) \left(1 - \frac{n(t)}{K} \right),$$

it quickly becomes an enormous headache (we have simplified the notation from that of eqn. 3.5a by using r rather than r_d for the intrinsic growth rate). If you specify the parameters and initial values of the variables, however, the recursion equation can be easily iterated using a calculator, a spreadsheet program, or a mathematical software package (Figure 4.2; see Lab exercise on the logistic model). This model has two parameters, r and K . The qualitative behavior of the model does not change much for different values of the carrying capacity K (we used $K = 1000$; see Problem 4.3), but it appears to be extremely sensitive to the value of the intrinsic growth rate r .

When the population starts below carrying capacity, it increases toward K , growing more rapidly for larger values of r . When r is small, the population size smoothly approaches the carrying capacity, following an S-shaped curve. When r is large enough, however, growth is so rapid that the population *overshoots* K . When r is slightly above two, the population size appears to cycle between two points. For example, when $r = 2.1$ (Figure 4.2a), the low point in the cycle is near 823. If we plug $K = 1000$, $r = 2.1$, and $n(t) = 823$ into the logistic

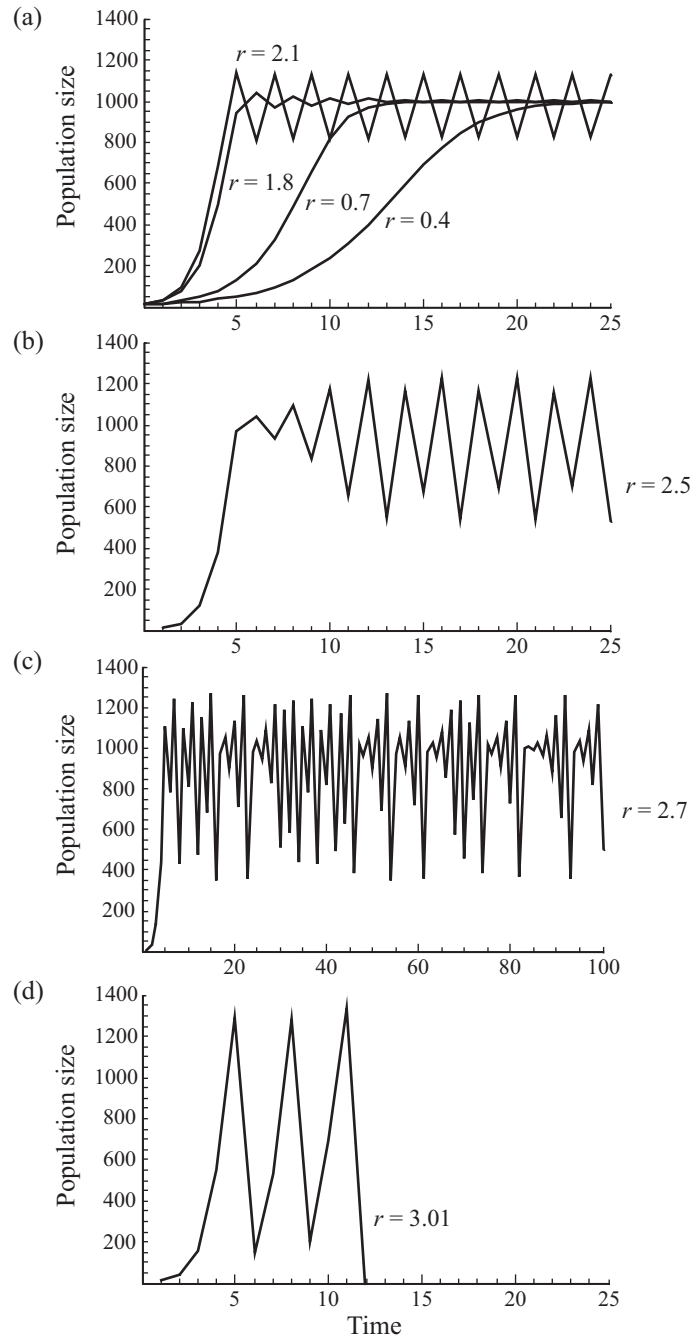


Figure 4.2: The dynamics of the logistic model in discrete time. The recursion equation (3.5a) is iterated for various values of the intrinsic growth rate, r . In each case, the initial population size is $n(0) = 10$ and the carrying capacity is $K = 1000$. More time steps are illustrated in Figure 4.2c to emphasize that the population dynamics do not follow a simple cycle when $r = 2.7$. Because time is discrete, the model only predicts the population sizes at integer values of time, but we connect the values of $n(t)$ with lines to clarify the behavior.

recursion, we find that the population size in the next generation is 1129, far above the carrying capacity. If we then plug $K = 1000$, $r = 2.1$, and $n(t) = 1129$ into the logistic recursion, we find that the population size plummets back to 823, and the two-point cycle begins again. As we increase the growth rate further, the number of points visited on the cycle increases from two (e.g., when $r = 2.1$; Figure 4.2a), to four (e.g., when $r = 2.5$; Figure 4.2b), to eight (e.g., when $r = 2.56$), to sixteen (e.g., when $r = 2.565$), etc. But for slightly larger values of

r (e.g., when $r = 2.7$; Figure 4.2c), something bizarre happens: the population size fluctuates all over the place, with no detectable pattern. The dynamics of the model have now entered the world of *chaos* (Box 4.1). Superficially, chaotic dynamics appear to be entirely erratic and random, but the remarkable thing is that they are, in fact, entirely predictable and deterministic. If we know the exact values of the parameters and variables, we can predict the population size at any future time by simply iterating the recursion equation. On the other hand, chaotic dynamics imply that the future state of the system will be extremely sensitive to imprecision in the calculations, e.g., due to rounding errors (Box 4.1).

Even when the variables are changing chaotically, they eventually settle down and remain within a specific range of values. As r increases, this range increases until, at $r = 3$, the range includes negative values of the population size (Problem 4.12). At this point the population fluctuates so wildly that extinction ensues (e.g., $r = 3.01$; Figure 4.2d). Thus, counterintuitively, populations with higher growth rates are not necessarily better protected from extinction (see Figure 4.1.2 in Box 4.1).

What values of r are typical of population growth? This question was addressed by Hassell et al. (1976) in a survey of population growth in 28 insect species. Of these species, 26 exhibited intrinsic growth rates for which the logistic recursion predicts that the population size will eventually settle down at the carrying capacity, one led to the prediction of a periodic cycle (the Colorado potato beetle), and only one led to the prediction of chaotic behavior (blowflies under laboratory conditions). This suggests that periodic cycles and chaotic fluctuations might not be particularly relevant to the population growth of many species, although biologists continue to debate the importance of chaos in natural populations (Dennis et al. 2001; Hastings et al. 1993; Turchin and Ellner 2000). Nevertheless, chaos theory has fundamentally changed the way scientists view the world, forcing us to recognize that seemingly disordered behavior can be generated from very orderly rules, such as that embodied by the logistic equation.

Do we also see cycling and chaos in the continuous-time model of logistic population growth? While it is possible to obtain the general solution for the differential equation (3.5c), which we could then plot (see Chapter 6), let us instead consider how to analyze the differential equation numerically. A straightforward numerical approach is to use the definition of a derivative to relate a differential equation to a discrete-time difference equation:

$$\frac{dn}{dt} = \lim_{\Delta t \rightarrow 0} \left[\frac{n(t + \Delta t) - n(t)}{\Delta t} \right].$$

Although it would be impossible to simulate an infinitely small time step, we can choose some very small time interval Δt to obtain an approximation for the differential equation:

$$\frac{dn}{dt} \approx \left[\frac{n(t + \Delta t) - n(t)}{\Delta t} \right]_{\text{for } \Delta t \text{ very small}}$$

Box 4.1: Chaos

The fact that deterministic equations can lead to seemingly unpredictable dynamics was first discovered by Edward Lorenz in 1961 (see Lorenz 1963). As a meteorologist working at MIT, Lorenz was developing computer simulations to predict weather patterns using a complex model involving twelve differential equations. One day, he took the predicted weather from a printout of a previous simulation and started a new set of simulations from these values, holding everything else constant. Soon thereafter, however, he noticed that the predicted weather was completely different from his previous results. He later realized that he started the second simulation at a very slightly different position because he had rounded a variable when printing it out. While the initial position was off by just 0.000127, the long-term weather predictions were entirely different. A defining characteristic of chaos is this *sensitivity to initial conditions*, such that two trajectories that start near one another grow apart over time until they are no nearer than two trajectories that started far apart. This sensitivity to initial conditions has become known as the “butterfly effect:”

The flapping of a single butterfly’s wing today produces a tiny change in the state of the atmosphere. Over a period of time, what the atmosphere actually does diverges from what it would have done. So, in a month’s time, a tornado that would have devastated the Indonesian coast doesn’t happen. Or maybe one that wasn’t going to happen, does. Stewart (1997), p. 141.

The emergence of chaos from entirely deterministic equations led Lorenz to conclude that there was little hope of predicting long-term weather patterns.

In 1974, the biologist Robert May (1974) published the simplest equation known to exhibit chaos: the logistic equation in discrete time. The logistic model involves a single variable and describes population growth as a quadratic function of the current population size (equation 3.5a). Because the population size can overshoot the equilibrium, chaotic fluctuations around the equilibrium are observed when the growth rate is large (e.g., $r = 2.7$ in Figure 4.2). Because the dynamics are chaotic, the dynamics are sensitive to initial conditions. For example, two populations whose initial sizes are very similar (e.g., 10,000 and 10,001) eventually become as different in population size as a population whose initial size is dramatically different (see Figure 4.1.1).

In the continuous-time logistic model, however, chaos is not observed (see Figure 4.3). Indeed, continuous-time models with only one or two variables never exhibit chaos. This does not mean that continuous-time models are always nicely behaved. In fact, continuous-time models with three or more variables can exhibit chaos. Indeed, chaotic dynamics are a common feature of food webs involving more than two species (Hastings and Powell 1991; Klebanoff and Hastings 1994; McCann and Hastings 1997). Interestingly, these models have been used to show that some forms of species interactions (e.g., linear food webs) are more prone to exhibit chaos than other forms of species interactions (e.g., omnivory).

We have mentioned that a defining feature of chaos is sensitivity to initial conditions. There are other clues that a model might exhibit chaos. One of them is that the system tends to oscillate between a number of states that doubles (“bifurcates”) repeatedly as a parameter in the model is altered. This sort of behavior is observed in the logistic model as we increase the intrinsic

(continued)

Box 4.1 (continued)

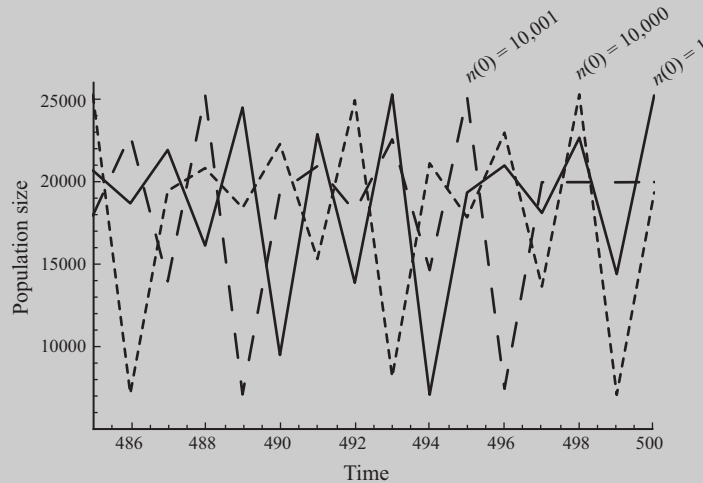


Figure 4.1.1: Sensitivity to initial conditions. A diagram of the population size versus time for the logistic equation (3.5a), starting from three different population sizes. After a few hundred generations, the population size dynamics of the two trajectories whose initial sizes were $n(0) = 10,000$ (short dashed lines) and $n(0) = 10,001$ (long dashed lines) are no closer together than they are to the population size dynamics starting from $n(0) = 1$ (solid lines). The parameters used were $r = 2.7$ and $K = 20,000$. Only time points 485 to 500 are shown to help see the differences between the trajectories.

growth rate, r (Figure 4.2). For low values of r , the population size approaches a single value K (e.g., $r = 0.7$ in Figure 4.2a). For higher values of r , the population size approaches an oscillation between two values, one above and one below K (e.g., $r = 2.1$ in Figure 4.2a). As r is increased further, the system settles down to a cycle involving four population sizes (e.g., $r = 2.5$ in Figure 4.2b). And as r is increased even further, the number of points through which the cycle passes (the *period*) doubles again and again. This period-doubling behavior is easiest to visualize using a *bifurcation diagram*.

Bifurcation diagrams illustrate the eventual states of a system on the vertical axis as a function of a parameter of interest on the horizontal axis. In the logistic model, the dynamics of the logistic model are quite sensitive to the intrinsic growth rate r , but not to other parameters such as the carrying capacity K (Problem 4.3). Thus, we use r as the parameter of interest in our bifurcation diagram (Figure 4.1.2). To produce this diagram, we iterated the recursion equation (3.5a) for a large number of generations (200) until the dynamics approached an equilibrium, a cycle, or showed no tendency to settle down. We then took the last 20 time points and plotted their values on the vertical axis. For small growth rates ($r < 2$), these last 20 time points were always very near the carrying capacity ($K = 1000$). The first period doubling occurred at $r = 2$, above which the population size cycled between two values (e.g., with $r = 2.1$, the population size

(continued)

Box 4.1 (continued)

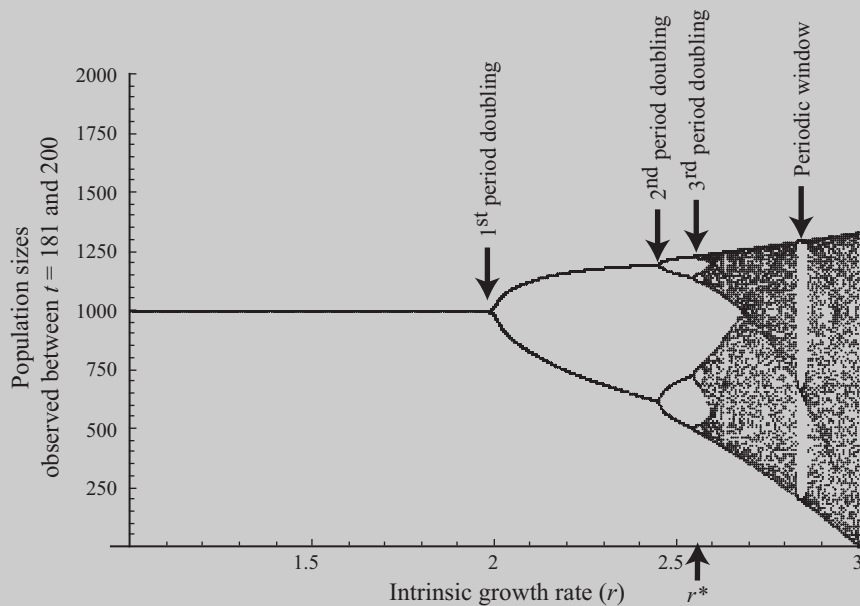


Figure 4.1.2: A bifurcation diagram for the logistic model. The recursion equation (3.5a) was iterated for 200 time steps, the last 20 of which are plotted on the vertical axis over a range of r values along the horizontal axis (r was increased from 1 to 3 in steps of 0.001). In every case, the initial population size was $n(0) = 10$, and the carrying capacity was $K = 1000$.

cycled between $n = 823$ and $n = 1129$, which give the vertical positions of the two points on the bifurcation diagram for $r = 2.1$). These two values grew further apart from one another until about $r = 2.45$, where the next period-doubling event occurred. Each subsequent period-doubling event occurred after shorter and shorter intervals in r . Eventually, the period doublings occurred so rapidly as r increased that the dynamics passed through a point at which an infinite number of period doublings had occurred. It is at this point that the dynamics become chaotic (at about $r^* = 2.569944$).

Period doubling is one route to chaos, and bifurcation diagrams allow us to visualize this process. If the bifurcation diagram for a model exhibits a forklike shape with tongs that divide faster and faster as the parameter of interest increases (as in Figure 4.1.2), expect to see chaos in the model.

A bifurcation diagram also illustrates that chaos does not imply a complete lack of order. In fact, for r values slightly above r^* , the population size remains within a fairly narrow region around the carrying capacity. This region expands as r increases and eventually includes zero at $r = 3$. Between the onset of chaos at r^* and extinction at $r = 3$, “periodic windows” appear, where the population size no longer fluctuates chaotically but cycles once again between a limited set of values. For example, a cycle among three points is observed at $r = 2.84$. Forklike bifurcations also occur within these periodic windows as r is increased further until, once again, the dynamics become chaotic.

Rearranging, we then get a recursion equation

$$n(t + \Delta t) = n(t) + \frac{dn}{dt} \Delta t,$$

which we can iterate, just as we did in Figure 4.2. As long as we choose a sufficiently small time interval Δt , the numerical solution for $n(t)$ should be consistent with the differential equation. By “consistent,” we mean that the slope of the curve of $n(t)$ versus t should equal equation (3.5c) for dn/dt at every point in time.

The above method is known as Euler’s method. Unfortunately, it is not very efficient because very small values of Δt are needed to obtain accurate predictions when n changes rapidly. Much more powerful and efficient algorithms have been developed to solve differential equations numerically (e.g., see Press 2002). Fortunately for us, mathematical software packages, including Maple, Matlab, and *Mathematica*, incorporate these algorithms “behind the scenes,” making it fairly painless to solve differential equations numerically. For example, *Mathematica* contains code that is over 500 pages long to obtain numerical solutions for differential equations that are both accurate and efficient; all we have to do is remember the command name: `NDSolve` for “Numerical Differential-equations Solve” (see Lab exercise on solving differential equations).

Figure 4.3 illustrates the numerical solution to the logistic differential equation for various values of the intrinsic growth rate r . Regardless of how large r gets, there is no sign of overshooting, cycles, chaos, or extinction! The continuous- and discrete-time formulations agree when the intrinsic growth rate is small (Figure 4.4), but they lead to completely different predictions when r is large. For example, when r is greater than 3, the discrete-time model predicts that the population will go extinct, while the continuous-time model predicts that the population rises quickly and steadily to carrying capacity, where it remains forever after.

What is going on? Conceptually, the reason for the discrepancy between discrete- and continuous-time models is that the population growth that accrues in the discrete-time model over one time step is based on the population

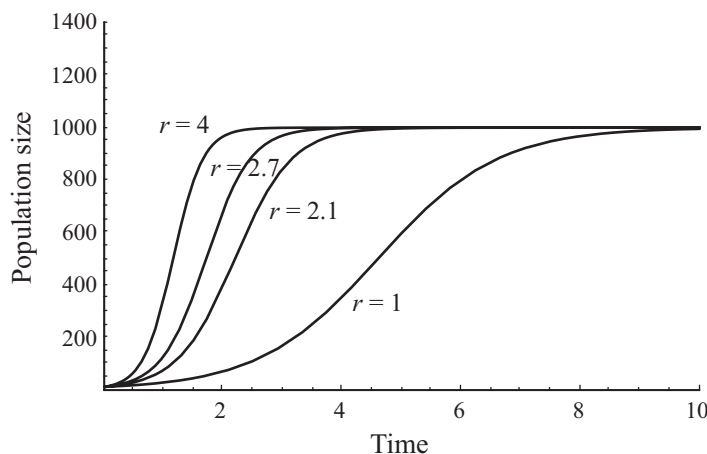


Figure 4.3: The dynamics of the logistic model in continuous time. The differential equation (3.5c) is solved numerically for various values of the intrinsic growth rate r using the command `NDSolve` in *Mathematica*. In each case, the initial population size is $n(0) = 10$ and the carrying capacity is $K = 1000$.

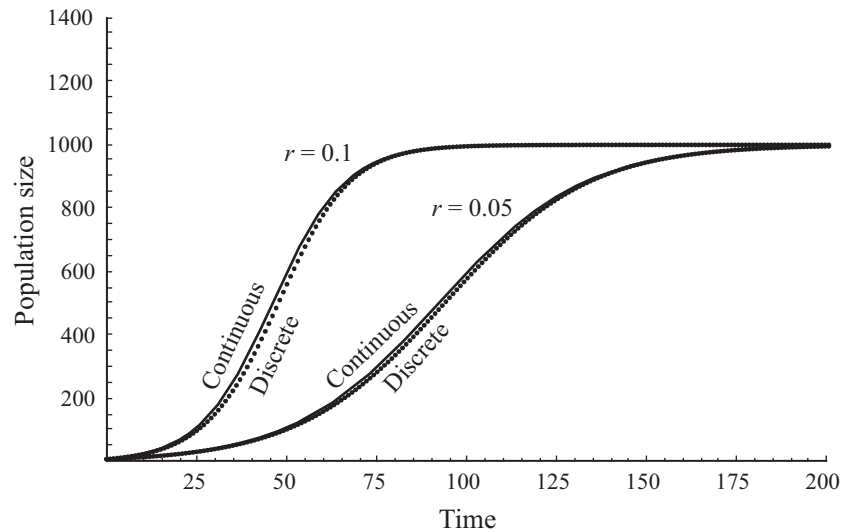


Figure 4.4: A comparison of the discrete-time and continuous-time logistic models. For low intrinsic growth rates, the two models predict similar trajectories for the population size over time. See Figures 4.2 and 4.3 for further details.

size at the beginning of the time step. There is nothing to prevent the population from growing past the carrying capacity by the end of the time step. In contrast, in the continuous-time model, the effect of population size feeds back continuously on the growth rate as the number of individuals changes. This instantaneous feedback prevents the population from overshooting and oscillating around its carrying capacity.

This sensitivity to the parameters and to the way that we model population growth must make us cautious in applying the logistic model uncritically to any natural population. To use the discrete-time model, we must be sure that the population satisfies the assumption that each event (e.g., births and deaths) happens only once per time unit and depends only on the preceding values of the variables. To use the continuous-time model, we must be sure that the population satisfies the assumption that each event (e.g., births and deaths) can happen in any order in any small interval of time, and that all individuals are equivalent, regardless of their age.

Other aspects of the logistic model are suspect and deserve to be critically evaluated. For example, we have assumed that the number of individuals can take on any real value, but in reality the number of individuals must be an integer. Furthermore, the number of offspring per parent will vary due to chance events (see Chapter 14). Making the logistic model more realistic in these ways causes dramatic changes to the predictions and eliminates the possibility of chaos (Henson et al. 2001).

At this point, you might be worried about the value of modeling if the results are so sensitive to the parameters and assumptions. Even when some predictions are sensitive, however, other predictions might not be. Results that do not depend very much on the parameters and underlying assumptions are

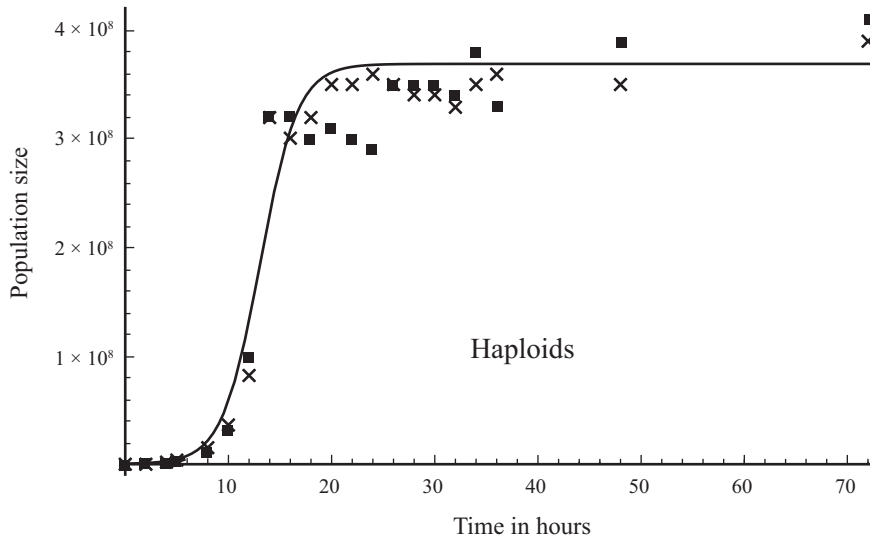


Figure 4.5: Dynamics of population growth in haploid yeast. The number of yeast in a flask is plotted over time using the consumer-resource model (3.17) where the resources are non-renewable ($\theta = 0$) and cell death is ignored ($d = 0$). The dynamics predicted by the continuous-time logistic model and the consumer-resource model lie exactly on top of each other, providing equally good fits to the data from Mable and Otto (2001) (crosses and squares, see Figure 3.4a). Parameter a was set to 1.47×10^{-9} ($= r_{\text{haploid}}/K_{\text{haploid}}$) and the initial resource level measured in terms of the number of cells that could be produced, $\varepsilon n_1(0)$, was set to 3.70×10^8 ($= K_{\text{haploid}} - n_2(0)$) (see Chapter 9). The fact that these models are mathematically equivalent explains why yeast in a vial display logistic growth, even though the resources are not constant over time.

called *robust*, and we tend to place more stock in such results. For example, as long as growth rates are not too large, populations grow according to an S-shaped curve in both discrete and continuous time (Figure 4.4). Furthermore, identical behavior is observed in a consumer-resource model in which we explicitly track the level of resources as well as the number of consumers (Figure 4.5; proven formally in Chapter 9). Thus, growth according to an S-shaped curve is a fairly robust prediction of the logistic model, breaking down only with high growth rates in a discrete-time model.

Second, even when a model is sensitive to assumptions, it can suggest new and interesting avenues of research. Perhaps the differences in model behavior that occur as a result of different assumptions provide an explanation for different population dynamics in nature. This might be addressed, for example, by determining if those species that persist despite having a high growth rate ($r > 3$) tend to replicate continuously over time.

4.2.3 The Dynamics of Models of Natural Selection

Not all models are as sensitive to the underlying assumptions as the logistic model. For example, the haploid model of natural selection displays very similar dynamics in both discrete and continuous time (Figure 4.6). The same is

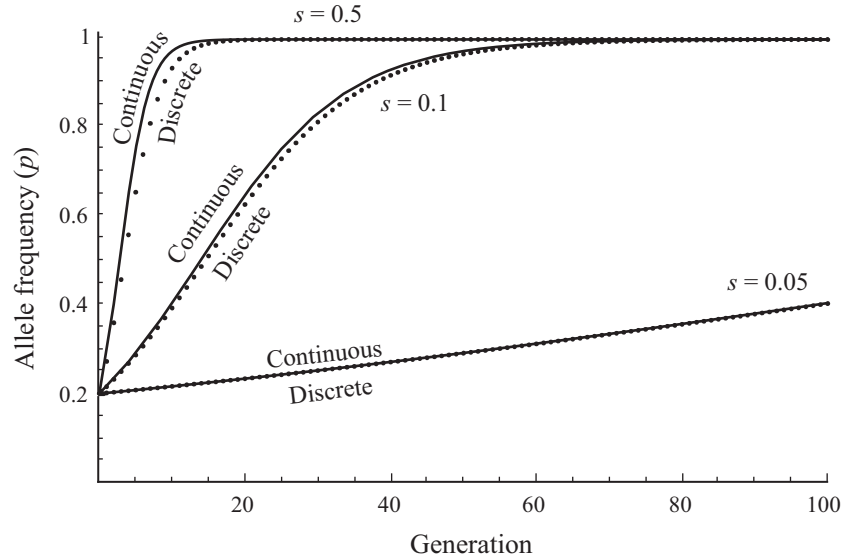


Figure 4.6: A comparison of the discrete-time and continuous-time models of haploid selection. For various values of the selection coefficient, s , the recursion equation (3.9) is iterated, and the differential equation (3.11b) is solved numerically using `NDSolve` in *Mathematica*. The predictions from the two models are in qualitative agreement even when selection is strong. The initial frequency of allele A is $p(0) = 0.2$.

true of the diploid model (not shown). As a result, both give a similar picture of the dynamics of natural selection.

Nevertheless, the behavior of models of selection is sensitive to the ordering of the fitnesses. In Figure 4.7, we plot the allele frequency as a function of time using the discrete-time, diploid model to gain some understanding of the effects of natural selection. These graphs appear to reveal three main types of behavior. If the heterozygotes have the highest fitness (e.g., $W_{AA} = 0.8$, $W_{Aa} = 1$, and $W_{aa} = 0.95$), natural selection tends to preserve both alleles (Figure 4.7a). Such selection is called “heterozygote advantage” or “overdominance.” In contrast, when heterozygotes have intermediate fitness (e.g., $W_{AA} = 1$, $W_{Aa} = 0.95$, and $W_{aa} = 0.8$), natural selection increases the frequency of one allele over the other (Figure 4.7b). This form of selection is called “directional selection.” Finally, when heterozygotes have the lowest fitness (e.g., $W_{AA} = 0.95$, $W_{Aa} = 0.8$, and $W_{aa} = 1$), natural selection favors different alleles depending on the initial allele frequency (Figure 4.7c). Such selection is called “heterozygote disadvantage” or “underdominance.” These results are restricted to the parameter values chosen, but the methods introduced in Chapter 5 will confirm that the behavior of the model always falls into one of the above three classes, depending on the ordering of the fitnesses.

4.2.4 The Dynamics of Competing Species

The above examples involve only a single variable. Similar plots can be drawn for models involving more than one variable, with different curves illustrating the dynamics of different variables. In discrete-time models, such plots

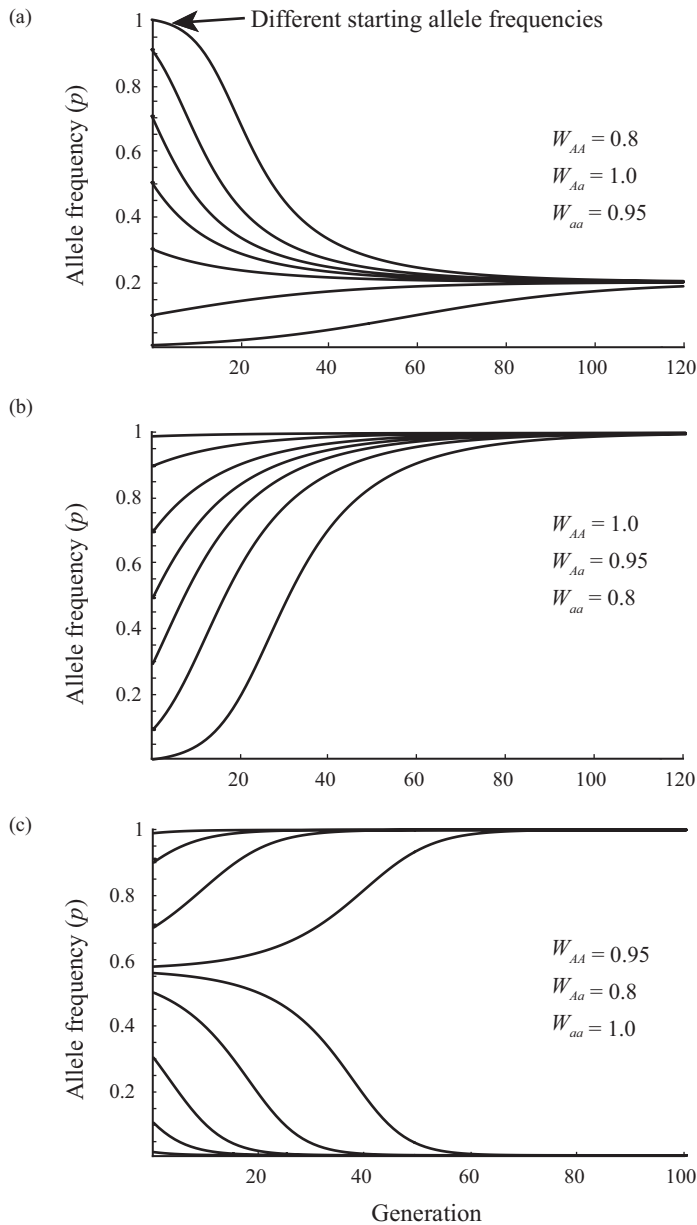
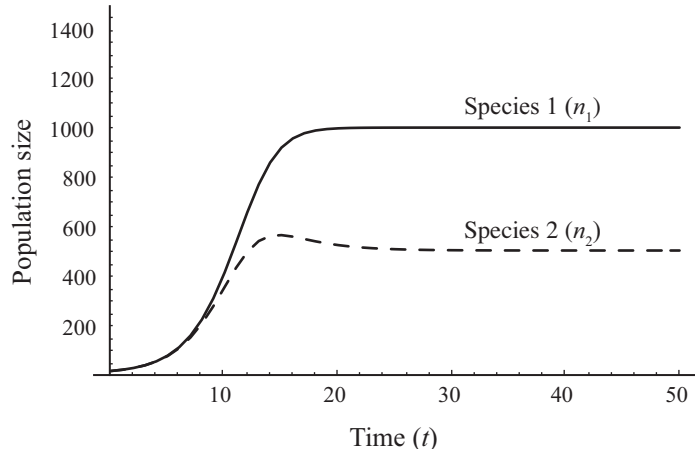


Figure 4.7: The discrete-time model of diploid selection. The recursion equation (3.13a) is iterated for various values of the initial frequency of allele A, $p(0) = 0.01, 0.1, 0.3, 0.5, 0.7, 0.9, 0.99$ (as indicated by the intercept with the y-axis). (a) Heterozygote advantage ($W_{AA} = 0.8$, $W_{Aa} = 1$, and $W_{aa} = 0.95$). (b) Directional selection favoring allele A ($W_{AA} = 1$, $W_{Aa} = 0.95$, and $W_{aa} = 0.8$). (c) Heterozygote disadvantage ($W_{AA} = 0.95$, $W_{Aa} = 0.8$, and $W_{aa} = 1$). Because time is discrete, the model only predicts the frequencies at integer values of time, but we connect the values of $p(t)$ with lines to clarify the behavior.

are again constructed by iteration of the recursion equations, where now each variable must be calculated in the next time step, as a function of all of the variables in the current time step.

For example, Figure 4.8 illustrates the dynamics of two species following the Lotka-Volterra equations with $r_1 = r_2 = 0.5$, $\alpha_{12} = 0$, $\alpha_{21} = 0.5$, and $K_1 = K_2 = 1000$. Under this (arbitrary) choice of parameters, species 1 competes for the resources of species 2 ($\alpha_{21} > 0$) while species 2 does not compete for the resources of species 1 ($\alpha_{12} = 0$). Starting from a population size of ten for each

Figure 4.8: The Lotka-Volterra model of competition in discrete time. The recursion equations (3.14a) are iterated, beginning from a population size of ten for each species ($n_1(0) = 10$, $n_2(0) = 10$). The parameters in the model were set to $r_1 = r_2 = 0.5$, $\alpha_{12} = 0$, $\alpha_{21} = 0.5$, and $K_1 = K_2 = 1000$. Because time is discrete, the model only predicts population sizes at integer values of time, but we connect these values with lines to clarify the behavior.



species, $n_1(0) = 10$, $n_2(0) = 10$, we can plug these parameters into equation (3.14) to predict the population size in the next time step:

$$\begin{aligned}
 n_1(1) &= n_1(0) + r_1 n_1(0) \left(1 - \frac{n_1(0) + \alpha_{12} n_2(0)}{K_1} \right) \\
 &= 10 + 0.5 \times 10 \left(1 - \frac{10 + 0 \times 10}{1000} \right) \\
 &= 14.95, \\
 n_2(1) &= n_2(0) + r_2 n_2(0) \left(1 - \frac{n_2(0) + \alpha_{21} n_1(0)}{K_2} \right) \\
 &= 10 + 0.5 \times 10 \left(1 - \frac{10 + 0.5 \times 10}{1000} \right) \\
 &= 14.925.
 \end{aligned}$$

After only one time step, the expected number of offspring is lower for species 2, not because it has a lower intrinsic growth rate or carrying capacity, but because it experiences more competition from the other species. This difference in growth continues, until eventually the system reaches a point where species 1 is near its carrying capacity, while species 2 sustains only half the number of individuals expected on the basis of its carrying capacity. For these parameters, a very similar graph is produced if we numerically solve the differential equations (3.15) for the continuous-time model of competition (Problem 4.10).

Variable versus variable plots illustrate the future state (or change) of a variable (vertical axis) versus the current state of the variable (horizontal axis).

4.3 Plots of Variables as a Function of the Variables Themselves

In the above graphs, we plotted the variable of interest over time (the independent variable). The second type of graph that we discuss is easier to generate; we simply plot the equation (recursion, difference, or differential) as a function of the variable itself. Such plots are particularly helpful in understanding the

dynamics of one-variable models because they clarify how the direction of change depends on the current state of the system.

4.3.1 Plotting the Value of a Variable at Time $t + 1$ versus at Time t

Consider the haploid model of natural selection with the recursion equation (3.9),

$$p(t + 1) = \frac{W_A p(t)}{W_A p(t) + W_a (1 - p(t))}.$$

Figure 4.9 plots $p(t + 1)$ as a function of $p(t)$ when W_A is greater than W_a (solid curve in Figure 4.9a) and when W_A is less than W_a (solid curve in Figure 4.9b). In both cases, we have also drawn a dashed diagonal 1:1 line. The diagonal line represents those special cases where $p(t + 1) = p(t)$. Wherever the recursion curve crosses the diagonal line, the allele frequency in the next time step $p(t + 1)$ will equal the allele frequency in the previous time step $p(t)$. Such a point is called an “equilibrium” because it remains unchanged over time (see Chapter 5). Thus, if the system starts at an equilibrium value for the allele frequency, it will remain there forevermore.

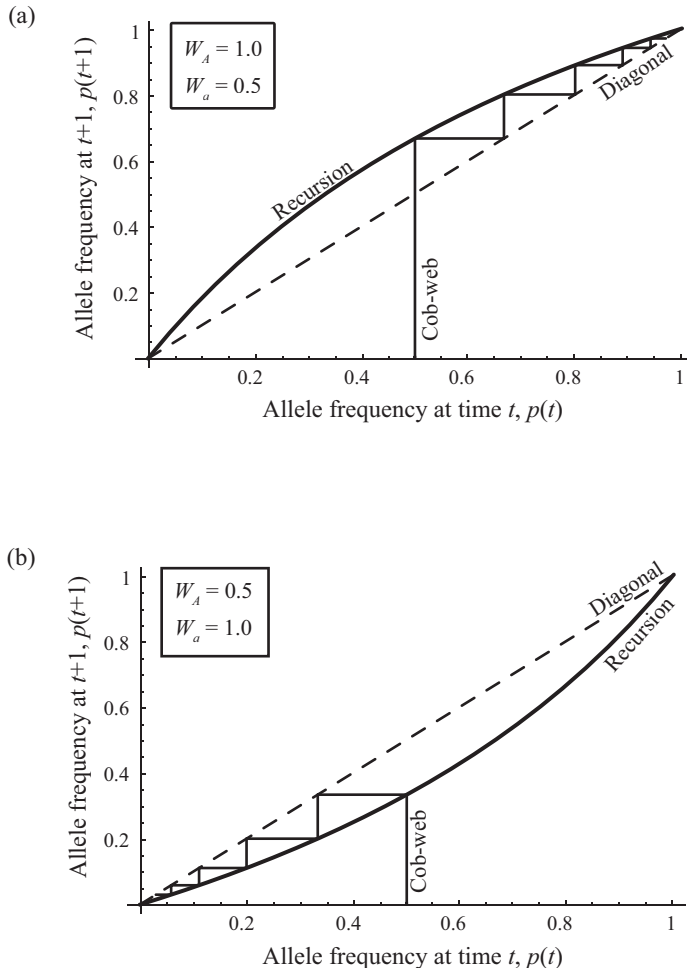


Figure 4.9: Allele frequency recursion in the haploid model of selection. The frequency of allele A at $t + 1$ is plotted against the frequency of allele A at t , using the recursion equation (3.9). The diagonal line (dashed) represents the case where $p(t + 1) = p(t)$. At any point where the recursion curve falls above the diagonal line, the allele frequency increases over time, as in (a) where $W_A = 1$ and $W_a = 0.5$. At any point where the recursion curve falls below the diagonal line, the allele frequency decreases, as in (b) where $W_A = 0.5$ and $W_a = 1$. The vertical and horizontal lines starting at $p(0) = 0.5$ illustrate “cobwebbing,” a procedure that can be used to determine changes to the variable over time (Box 4.2).

In the haploid model, there are two places where the recursion function crosses the diagonal: when the frequency of A is zero and when it is one (Figure 4.9). Again, we know that this is true only for the particular fitnesses used in the figure, but we will see in Chapter 5 that these two points represent the only two equilibria of the haploid model with constant fitnesses.

Recipe 4.1

Identifying Equilibria Graphically

A system at “*equilibrium*” does not change over time.

- In a plot of $n(t + 1)$ versus $n(t)$, any point where the recursion equation crosses the diagonal line represents an equilibrium, because $n(t + 1) = n(t)$.
- In a plot of Δn versus $n(t)$, any point where the difference equation crosses the horizontal axis represents an equilibrium, because $\Delta n = 0$.
- In a plot of dn/dt versus $n(t)$, any point where the differential equation crosses the horizontal axis represents an equilibrium, because $dn/dt = 0$.

Now, focus your attention on whether the recursion lies above or below the diagonal. Before reading on, think about what this means. Whenever the recursion lies above the diagonal, $p(t + 1)$ is greater than $p(t)$, and allele A increases in frequency. For the parameter values in Figure 4.9a (where $W_A > W_a$), the recursion is above the diagonal for all values of the allele frequency ($0 < p(t) < 1$). As a result, allele A rises in frequency every generation until it approaches one (“fixation”). Conversely, whenever the recursion falls below the diagonal (as in Figure 4.9b, where $W_A < W_a$), $p(t + 1)$ is always lower than $p(t)$. As a result, allele A decreases in frequency until it is lost from the population. You can even use these figures to run your own iterations by hand using a technique known as “cobwebbing” (Figure 4.9; see Box 4.2).

Recipe 4.2

Identifying Direction of Change Graphically

- In a plot of $n(t + 1)$ versus $n(t)$, if the recursion equation lies above the diagonal line, the variable increases over time ($n(t + 1) > n(t)$) and vice versa.
- In a plot of Δn versus $n(t)$, if the difference equation lies above the horizontal line, the variable increases over time ($\Delta n > 0$) and vice versa.
- In a plot of dn/dt versus $n(t)$, if the differential equation lies above the horizontal line, the variable increases over time ($dn/dt > 0$) and vice versa.

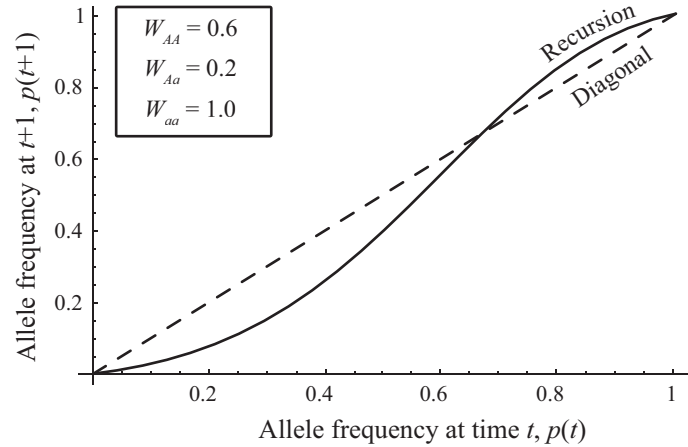
There are two general types of equilibria: *attracting* and *repelling*. As these categories suggest, variables tend to move toward attracting equilibria over time, whereas they tend to move away from repelling equilibria over time. Attracting states are also known as *stable* equilibria, while repelling states are known as *unstable* equilibria. As an example, imagine trying to balance on a bicycle whose pedals are locked in place. If you could manage to start perfectly upright and remain completely motionless, then in the absence of any external forces (e.g., a gust of wind), you would remain upright indefinitely. Starting slightly off vertical, however, you would fall either to the left or to the right. Measuring the angle of the bike from the horizontal position, the bike has three equilibrium states: lying flat on its left-hand side (0°), lying flat on its right-hand side (180°), and standing perfectly upright (90°). The first two equilibria (0° and 180°) are both attracting, or stable, equilibria because the bicycle will fall to the ground once it has started to tip over. The third equilibrium (90°) is repelling, or unstable, because the bicycle will fall over once it starts to tip away from this point.

Stable equilibria are subdivided further into *locally stable* equilibria, which are attracting from nearby states, and *globally stable* equilibria, which are attracting from all states. There are two locally stable equilibria in the bicycle example: lying on the left or lying on the right. The bicycle can end up in either state depending on the direction in which it starts to fall. If there were a person on the right pushing the bike back if it ever started to fall, then the bicycle would only have two equilibria, an unstable equilibrium in the upright position (i.e., 90°) and a stable equilibrium on its left-hand side (0°). The latter equilibrium is globally stable because it is attracting for all possible starting conditions (except for the perfectly balanced upright position).

Returning to the haploid model of selection illustrated in Figure 4.9a (where $W_A > W_a$), the frequency of allele A always rises toward $p = 1$ because the recursion lies above the diagonal. Therefore, $p = 1$ is a globally stable equilibrium. On the other hand, $p = 0$ is unstable, and the population moves further and further away from this point. The opposite is true in Figure 4.9b (confirm to yourself that $p = 1$ is unstable and $p = 0$ is globally stable).

If an equilibrium is globally stable, it must also be locally stable, but the opposite is not true. Consider the diploid model of natural selection when the heterozygote is least fit. Figure 4.10 shows $p(t + 1) = p(t)^2 W_{AA}/\bar{W} + p(t) q(t) W_{Aa}/\bar{W}$ plotted against $p(t)$ for the case $W_{AA} = 0.6$, $W_{Aa} = 0.2$, and $W_{aa} = 1$. Judging from where the recursion curve crosses the diagonal line, there are now three equilibria ($p = 0$, $p = 1$, and $p = 0.66$). For small initial allele frequencies ($0 < p < 0.66$), the recursion lies below the diagonal, implying that the allele frequency decreases in each generation and approaches $p = 0$. Therefore $p = 0$ is a locally stable equilibrium. For high initial allele frequencies ($0.66 < p < 1$), the recursion lies above the diagonal, implying that the allele frequency increases in each generation and approaches $p = 1$. Therefore, $p = 1$ is also a locally stable equilibrium. Thus, there is a region (known as a *basin of attraction*) within which p moves toward 0, and another region within which p moves toward 1. As a result, neither of these equilibria is globally stable (see Chapter 5).

Figure 4.10: Allele frequency recursion in the diploid model of selection. The frequency of allele A at $t + 1$ is plotted against the frequency of allele A at t , using the recursion equation (3.13a). The diagonal line (dashed) represents the case where $p(t + 1) = p(t)$. The allele frequency decreases over time when $p(t)$ is less than 0.66 (recursion lies below the diagonal) but increases over time when $p(t)$ is greater than 0.66 (recursion lies above the diagonal). Here, $W_{AA} = 0.6$, $W_{Aa} = 0.2$, and $W_{aa} = 1$.



We will return to the issue of stability in greater depth in Chapter 5. For the time being, however, we can foreshadow a key result by thinking about the feature of these graphs that determines whether an equilibrium is locally stable or unstable. For an equilibrium to be locally stable, the variable must rise in frequency when it begins below the equilibrium (i.e., the recursion equation must lie above the diagonal), and the variable must decline in frequency if it begins above the equilibrium (i.e., the recursion equation must fall below the diagonal). Considering that the diagonal line has a slope of one (it is the 1:1 line), this implies that the slope of the recursion at the equilibrium must therefore be less than one for the equilibrium to be locally stable. Conversely, if the slope of the recursion is greater than one at an equilibrium, then the equilibrium will be unstable. Check to see that this claim is valid for the equilibria in Figures 4.9 and 4.10. The haploid and diploid models of selection are good examples to begin to think about stability, because the slope of the recursion with respect to the variable p is always positive (Problems 4.5 and 4.6). This is not true of all models, however, and more complicated dynamics arise when the slope is negative (Box 4.2).

4.3.2 Plotting the Change in a Variable versus the Variable at time t

Let us now consider a different type of plot, where instead of plotting the value of the variable in the next generation on the vertical axis, we plot the *change* in the variable on the vertical axis. Again we use the current value of the variable on the horizontal axis. The change in a variable is given by a difference equation in a discrete-time model or by a differential equation in a continuous-time model.

We first revisit the haploid model of selection. With weak selection, the change in the frequency of allele A is approximately $sp(t)(1-p(t))$ according to either the difference equation (3.10) for Δp or the differential equation (3.11b) for dp/dt (Chapter 3). In Figure 4.11, the rate of change, $sp(t)(1-p(t))$, is plotted as a function of $p(t)$. The population is at equilibrium if there is no change in

Box 4.2: Cobweb Plots of the Logistic Equation

Cobweb plots provide a graphical way of iterating a recursion equation involving a single variable. In a cobweb plot, the value of the variable at time $t + 1$ is plotted on the vertical axis as a function of the variable at time t along the horizontal axis (as in section 4.3). Starting from any initial position along the horizontal axis, the value of the variable in the next time step is determined by drawing a vertical line up to the curve for the recursion equation. The predicted value of the variable equals the height of the curve at this point. This value can then be used as the initial state of the system in the next time step. This procedure may be repeated for as many time steps as desired.

The iteration procedure is made easier by adding a diagonal line to the plot (i.e., the line where $n(t + 1) = n(t)$) and employing the method illustrated in Figure 4.2.1 for the logistic equation (3.5a). At step (a), the variable changes from an initial value of $n(0) = 400$ to $n(1) = 568$, which is determined by drawing a vertical line starting at $n(0)$ up to the recursion curve. (b) The starting value of the variable in the next time step is given by the height of this recursion curve, $n(1) = 568$. To use this height as the starting value in the next generation, we draw a horizontal line from the recursion curve to the diagonal line, which effectively sets the starting position in the next generation to 568, the final position in the previous generation (see Figure 4.2.1). (c) At this point, a vertical line is drawn to the recursion equation again, giving the variable in the next time step, $n(2) = 740$. From this point on, each time step corresponds to drawing a horizontal line from the recursion curve to the diagonal to obtain the starting value (d), then drawing a vertical line from the diagonal to the recursion curve (e) to determine the value of the variable in the next time step. This cobwebbing procedure can be used to predict the state of the variables over multiple time steps (Figure 4.2.1).

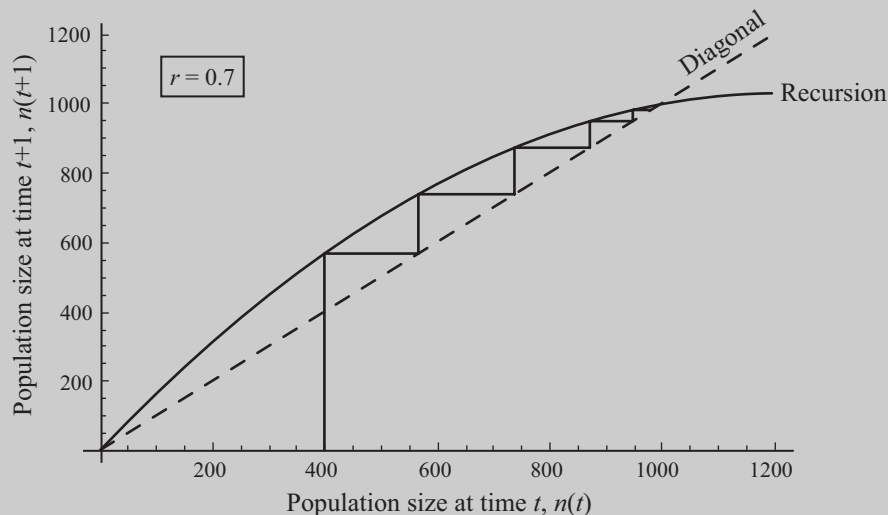


Figure 4.2.1: A cobweb diagram for the logistic model (low r). The starting condition and parameters of the logistic equation were $n(0) = 400$, $r = 0.7$, and $K = 1000$.

(continued)

Box 4.2 (continued)

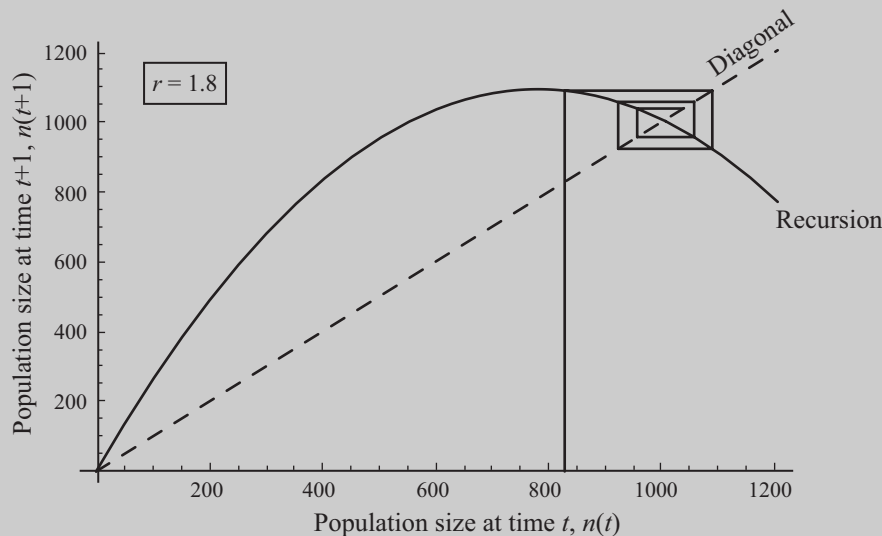


Figure 4.2.2: A cobweb diagram for the logistic model (intermediate r). The starting condition and parameters of the logistic equation were $n(0) = 825$, $r = 1.8$, and $K = 1000$.

In Figure 4.2.1, the intrinsic growth rate is low ($r = 0.7$), and the cobwebbing procedure suggests a smooth approach to the carrying capacity at $K = 1000$ (see Figure 4.2a). Overshooting the equilibrium does not occur in this case. As we increase r , however, the height of the recursion rises until it eventually lies above K when the system starts near to, but below, K (Figure 4.2.2 for $r = 1.8$). For example, when starting with an initial population size of $n(0) = 825$ (far left vertical line), the population grows to 1085 individuals, above the carrying capacity at $K = 1000$. Foreshadowing Chapter 5, the key feature of figures like Figure 4.2.2, where overshooting an equilibrium is possible, is that the slope of the recursion equation at the equilibrium of interest is negative (here, the slope at K is -0.8). This is mathematically equivalent to the statement that it is possible to rise above K starting at points slightly below K .

Even in Figure 4.2.2, the system approaches the carrying capacity (i.e., the cobwebs are moving toward the equilibrium). If r is increased even further, however, there comes a point where the height of the recursion equation rises so much that populations starting slightly below K grow so much that they end up further away from K on the other side (Figure 4.2.3 for $r = 2.5$). For example, when starting with an initial population size of $n(0) = 905$ (below the carrying capacity by 95 individuals), the population grows to 1186 individuals (above the carrying capacity by 186 individuals). The key feature of figures like Figure 4.2.3, where a system started near an equilibrium overshoots the equilibrium by so much that it moves further from the equilibrium, is that the slope of the recursion equation at the equilibrium of interest is less than -1 (here, the slope at K is -1.5). In Chapter 5, we will see that the insights provided by these cobweb plots are consistent with analytical methods that determine when an equilibrium is stable or unstable.

(continued)

Box 4.2 (continued)

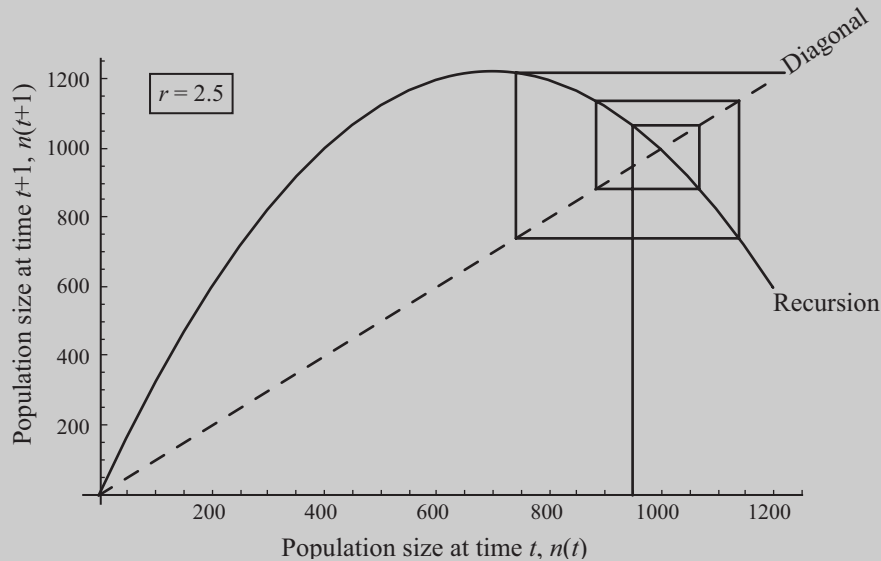


Figure 4.2.3: A cobweb diagram for the logistic model (high r). The starting condition and parameters of the logistic equation (3.5a) were $n(0) = 950$, $r = 2.5$, and $K = 1000$.

the variable, which occurs wherever the curve of the difference or differential equation crosses the horizontal axis (Recipe 4.1). Once again, we find equilibria at $p = 0$ and $p = 1$. As mentioned in Recipe 4.2, whenever the change in p is positive (i.e., lies above the horizontal axis), allele A increases in frequency (as in Figure 4.11a). This is consistent with the rise in allele frequency observed in Figure 4.6, where we plotted the allele frequency as a function of time. Similarly, whenever the change in p is negative, allele A decreases in frequency (as in Figure 4.11b).

Again, we can use these figures to gain some insight into the conditions that must be met for an equilibrium to be locally stable. Look at Figure 4.11 and think about how the stability of an equilibrium depends on the slope of the difference or differential equation. If the slope is positive at an equilibrium, then a variable that begins slightly below the equilibrium will decrease away from it (change is negative), while a variable that begins slightly above the equilibrium will increase away from it (change is positive). Thus, when the slope is positive, an equilibrium is unstable.

Conversely, if the slope is negative at an equilibrium, then a variable that begins slightly below the equilibrium will increase (change is positive), and a variable that begins slightly above the equilibrium will decrease (change is negative), so the equilibrium might be stable. We've waffled a bit, here, by saying "might be," because, for discrete-time models, it is possible for the variable to move toward the equilibrium, but overshoot it and end up even further away

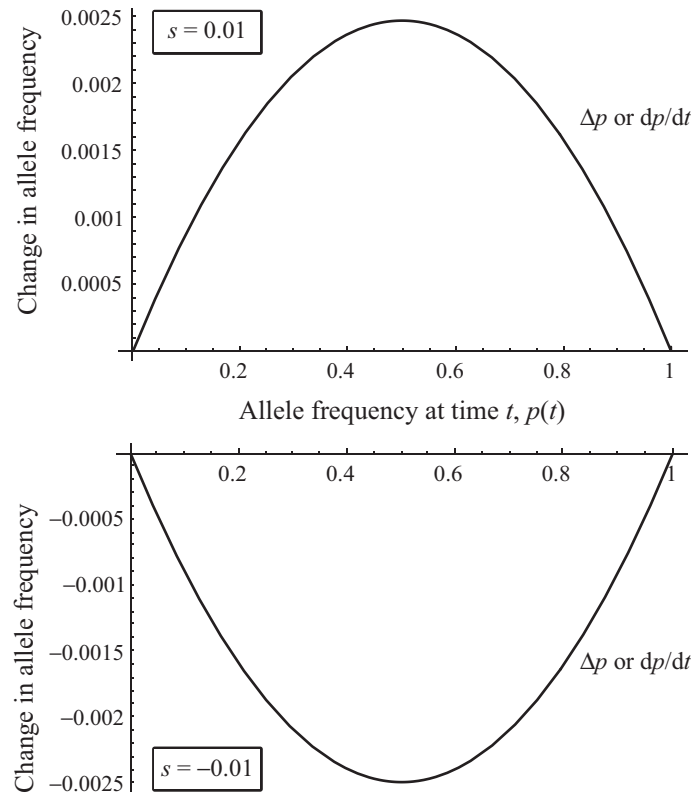


Figure 4.11: Change in allele frequency as a function of the current allele frequency in the haploid model of selection. The vertical axis describes the change in allele frequency, $\Delta p \approx dp/dt = s p(t)(1-p(t))$. The horizontal axis gives the current allele frequency $p(t)$. At any point where the curve falls above the horizontal axis, the allele frequency increases over time, as occurs in (a) for $s = 0.01$. At any point where the curve falls below the horizontal axis, the allele frequency decreases over time, as occurs in (b) for $s = -0.01$.

on the other side. We will address this issue in much greater detail in the next chapter, but for now it is enough to recognize that the slope of a difference or differential equation when plotted against the variable of interest provides important information about the stability of an equilibrium.

As another example, consider the logistic model of population growth. The change in population size is $rn(t)(1 - n(t)/K)$ according to either the difference equation (3.5b) for Δn or the differential equation (3.5c) for dn/dt (Chapter 3). Figure 4.12 plots this change as a function of $n(t)$ for $r = 1$ and $r = 2.5$, assuming a carrying capacity of 1000. In both cases, the function crosses the horizontal axis at $n = 0$ and $n = K$, which correspond to the two equilibria of the model. For both $r = 1$ and $r = 2.5$, if the population size begins below the carrying capacity it grows, while if it begins above the carrying capacity it shrinks. This suggests that the equilibrium $n = 0$ is unstable and that the $n = K$ equilibrium is stable. But we must be careful! In a discrete-time model, overshooting the equilibrium is possible (Figure 4.2). Thus, even though the population size moves in the direction of an equilibrium this does not mean that it gets closer to the equilibrium. For example, when the population begins at $n(0) = 900$, the change in population size is +225 for $r = 2.5$ (check by using the difference equation or by reading off the value from the graph). The population at the next time is thus $n(1) = 900 + 225 = 1125$, which is even further away from the equilibrium on the other side. Indeed, as we shall see in Chapter 5,

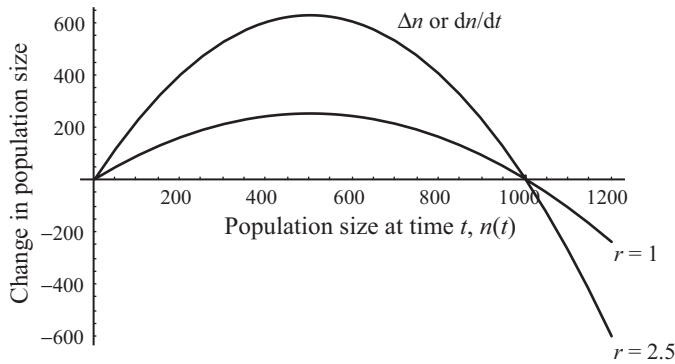


Figure 4.12: Changes in population size as a function of the current population size in the logistic model. The vertical axis describes the change in population size $\Delta n = dn/dt = r n(t) (1 - n(t)/K)$. The horizontal axis gives the current population size $n(t)$. At any point where the curve lies above the horizontal axis, the population size increases over time, as occurs below the carrying capacity of $K = 1000$. At any point where the curve falls below the horizontal axis, the population size decreases over time, as occurs above the carrying capacity of $K = 1000$. Two curves are shown, assuming $r = 1$ and $r = 2.5$.

the carrying capacity is not a stable equilibrium of the discrete-time model when the intrinsic growth rate is too high.

The problem of overshooting the equilibrium does not arise in the continuous-time logistic model. In a continuous-time model, a variable rises smoothly over time, passing through all possible points between where it begins and where it ends. For a variable to overshoot an equilibrium (e.g., at $K = 1000$), it would have to pass through the equilibrium. For example, for the population size to change smoothly from $n(0) = 900$ to $n(1) = 1125$ in a continuous-time model would require that the population size equal 1000 at some intermediate point in time. But when the variable hits the equilibrium, the change in the variable will equal zero (by the definition of an equilibrium), and the variable will not move any further. This provides another perspective on why the behavior of the logistic model in continuous time is much more tame than in discrete time. The logistic model, like any other continuous-time model with only a single variable, cannot exhibit overshooting or chaos because the variable cannot pass through an equilibrium without getting stuck.

The graphs considered in this section are less natural for models with more than one variable. The reason is that one cannot predict the state of the system in the next time step as a function of only one variable. Rather, one must know the current state of all variables to predict their future states. For models with two variables, you could draw a three-dimensional graph where the two axes in the plane (x and y) correspond to the current states of the two variables, and the vertical axis (z) corresponds to the state of the first variable in the next time step. But such a graph is not informative on its own. The first variable might reach a state where it does not change (i.e., the recursion equation may fall on the diagonal where $z = x$), but this is based on the current value of the second variable, which might still be changing. A much clearer picture emerges from phase-plane diagrams, introduced in the next section.

4.4 Multiple Variables and Phase-Plane Diagrams

In this third type of graph, each axis represents a different dependent variable. Time is represented as a series of dots, one for each time step, or as arrows pointing in the direction of change. Such graphs are called “phase-plane

diagrams.” Phase-plane diagrams are particularly helpful in understanding the dynamics of two-variable models because they indicate how each variable changes as a function of the other variable.

4.4.1 Phase-Line Diagrams

Before describing phase-plane diagrams for two-variable models, let us start with the simpler case of phase-line diagrams in single-variable models. The idea is to draw a line or axis spanning the range of possible values of the variable. At several points along the line, arrows are drawn that indicate the direction in which the variable moves. Figure 4.13 illustrates phase-line diagrams for the haploid and diploid models considered earlier. These phase-line diagrams

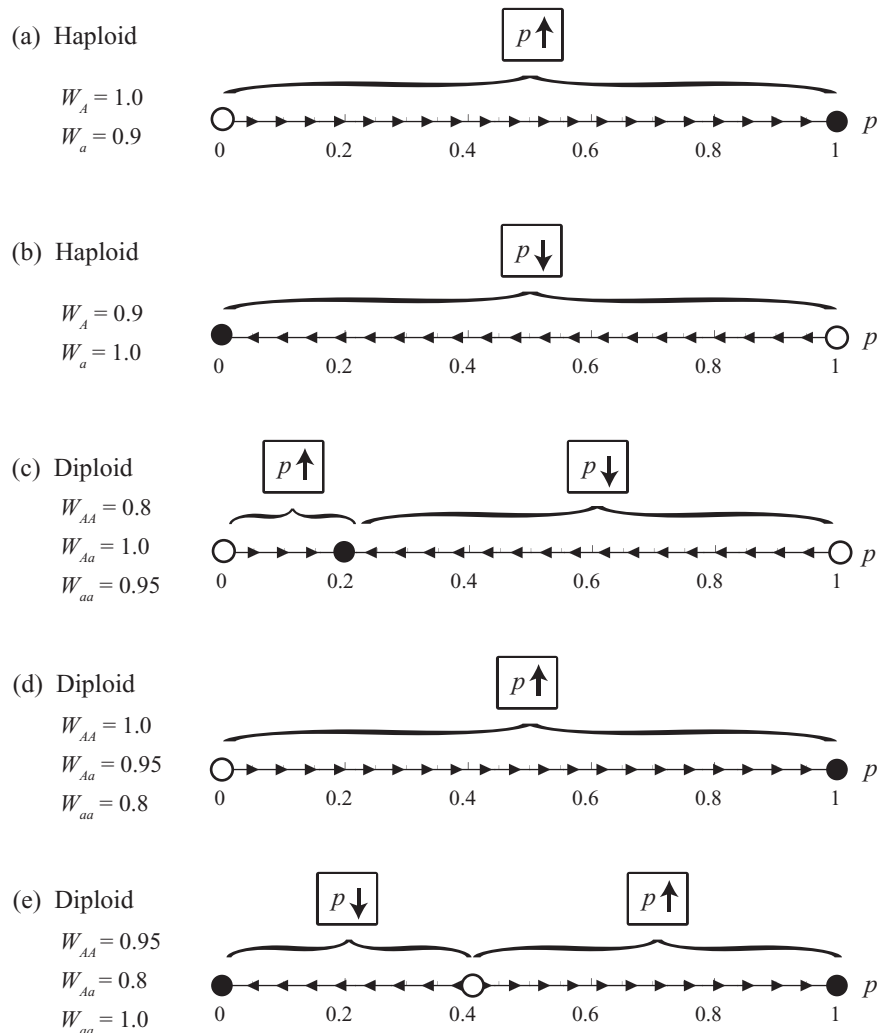


Figure 4.13: Changes in allele frequency illustrated on a phase-line diagram. Arrows illustrate the direction in which allele frequencies evolve. Arrows point toward solid circles and away from hollow circles.

provide a nice visual summary of the behavior of a model. At a glance, it is possible to see the direction in which a variable changes for a particular set of parameters.

In the haploid model of selection, the direction of the arrows is given by the sign of the difference equation (3.10) in discrete time:

$$\Delta p = \frac{s_d p(t)(1 - p(t))}{1 + s_d p(t)},$$

or the sign of the differential equation (3.11b) in continuous time:

$$\frac{dp}{dt} = s_c p(t)(1 - p(t)).$$

In either case, the direction of the arrows depends only on the sign of the selection coefficient because $p(t)(1 - p(t))$ is always positive (we ignore the case where $p(t)(1 - p(t))$ equals zero, which implies that one allele is absent), as must be the mean fitness in the denominator of Δp . Thus, when $s > 0$, the arrows always point to the right, and the allele frequency increases. We can conclude that $p = 1$ is globally stable whenever $s > 0$, regardless of the exact values of the selection coefficient or the initial conditions. Similarly, when $s < 0$, the arrows always point to the left, and $p = 0$ is globally stable.

In continuous-time models, phase-line diagrams tell us immediately which equilibria are locally stable (arrows point in) and which are unstable (arrows point out). In discrete-time models, however, we have to be careful in interpreting phase-line diagrams whenever it is possible for the system to overshoot the equilibrium. In the haploid model of selection, overshooting is not a concern, because it would require one of the allele frequencies to become negative, which is not possible (because the fitnesses cannot be negative, equation 3.8c never becomes negative). But, in general, it is difficult to tell from phase-line diagrams in discrete time whether movement along the arrows might lead a variable further away from an equilibrium on the other side.

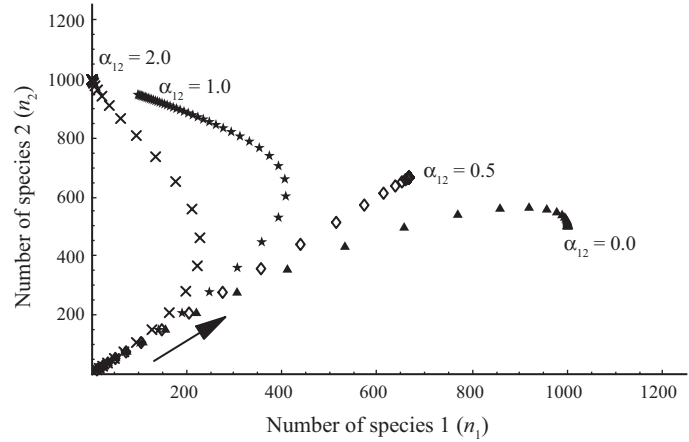
4.4.2 Phase-Plane Diagrams

We can extend phase-line diagrams to models with two variables using phase-plane diagrams, with the horizontal axis representing the range of one variable and the vertical axis representing the range of the other variable. On a phase-plane diagram, we can trace how the variables change over time, starting from a particular value along the horizontal and vertical axes. This is best illustrated by example.

In Figure 4.14, the dynamics of two competing species in the Lotka-Volterra model is illustrated by a phase-plane diagram. Starting from $n_1(0) = 10, n_2(0) = 10$, the recursion equations (3.14) were iterated for 50 time steps. At time step t , the population sizes were used to plot a point whose horizontal coordinate is the number of species 1 and vertical coordinate is the number of species 2. Four simulations were conducted using different competitive effects of species 2 on

A phase-plane diagram illustrates the state of one variable (vertical axis) versus the state of a second variable (horizontal axis).

Figure 4.14: Dynamics of the two-species Lotka-Volterra model of competition. The recursion equations (3.14) were iterated for 50 time steps starting with ten individuals of each species, $n_1(0) = n_2(0) = 10$. This was repeated four times using different competitive effects of species 2 on species 1: $\alpha_{12} = 0$ (triangles), $\alpha_{12} = 0.5$ (diamonds), $\alpha_{12} = 1$ (stars), and $\alpha_{12} = 2$ ($\times$). The remaining parameters were: $r_1 = r_2 = 0.5$, $\alpha_{21} = 0.5$, and $K_1 = K_2 = 1000$.



species 1: $\alpha_{12} = 0$ (triangles), $\alpha_{12} = 0.5$ (diamonds), $\alpha_{12} = 1$ (stars), and $\alpha_{12} = 2$ ($\times$), holding all other parameters constant ($r_1 = r_2 = 0.5$, $\alpha_{21} = 0.5$, $K_1 = K_2 = 1000$, $n_1(0) = n_2(0) = 10$). An arrow is added to the graph to show the direction in which time flows. For example, we saw above that, after one time step, the population sizes become $n_1(1) = 14.95$ and $n_2(1) = 14.925$ with $\alpha_{12} = 0$. Thus, in Figure 4.14, the $\{x, y\}$ coordinates of the first triangle corresponding to $t = 0$ are placed at $\{10, 10\}$, and the $\{x, y\}$ coordinates of the second triangle corresponding to $t = 1$ are placed at $\{14.95, 14.925\}$. These values of the variables are then plugged back into the recursion to get the coordinates at $t = 2, 3, 4$, etc. up until 50 time steps. This process is then repeated for the next value of α_{12} starting back at the initial point $\{10, 10\}$.

The population sizes grow in the same direction at first regardless of the value of α_{12} (along the arrow on Figure 4.14) because there are few competitors around. As the population sizes get larger, competition begins to play a more decisive role, causing the trajectories to diverge for different values of α_{12} . When species 2 has no effect on species 1 ($\alpha_{12} = 0$), the trajectory approaches the point $\{1000, 500\}$ after 50 times steps (filled triangles), as seen in Figure 4.8. For larger values of α_{12} , competition limits the number of species 1, shifting the trajectories to the left in Figure 4.14. Indeed, for $\alpha_{12} = 2$, species 2 eventually outcompetes species 1, which is driven to extinction ($\times$'s). In Figure 4.14, the symbols start piling on top of one another toward the end of the 50 time steps, implying that smaller and smaller changes occur per time step. This is a hallmark feature of variables as they near an equilibrium.

As a good check that the iterations have been coded correctly, consider the case with $\alpha_{12} = 0.5$ (diamonds). In this case, the parameters for species 1 and 2 are identical: $r_1 = r_2$, $\alpha_{12} = \alpha_{21}$, $K_1 = K_2$, and $n_1(0) = n_2(0)$. With such symmetry, we expect the number of each species to be the same every generation. Figure 4.14 shows this to be true, as the diamonds fall along the diagonal line.

4.4.3 Vector-Field Plots

A drawback of the above phase-plane plots is that they illustrate how the variables change only along certain trajectories (e.g., starting at $n_1(0) = n_2(0) = 10$). A more complete visual representation of the dynamics is possible by

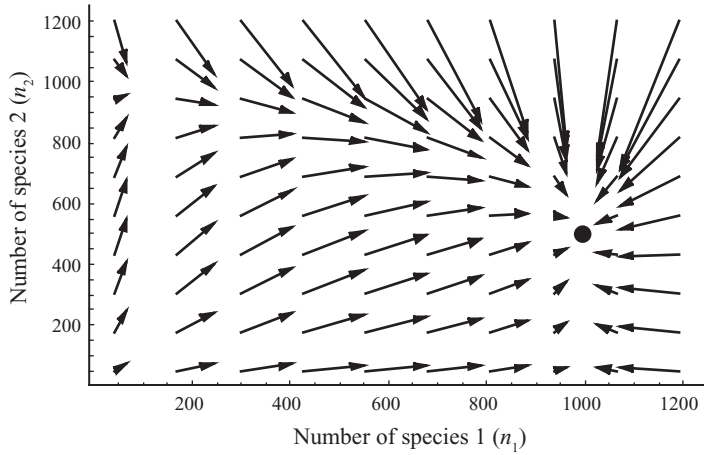


Figure 4.15: Vector-field plot for the two-species Lotka-Volterra model of competition. The change over one generation is shown from a series of different starting positions. All of the parameters are the same as in Figure 4.14 with α_{12} set to 0. The filled circle represents an equilibrium of this model ($n_1 = 1000$; $n_2 = 500$).

drawing arrows specifying the direction of change from a variety of different positions. Such a phase-plane diagram is known as a *vector-field plot*. Figure 4.15 is an example of a vector-field plot for the discrete-time model of competition, where each arrow represents the change over a time step in the numbers of species 1 (along the horizontal axis) and species 2 (along the vertical axis). To see how the arrows are drawn, consider the top left arrow. The base of this arrow is at $\{50, 1200\}$, which represents the starting population sizes of the two species. Plugging in these initial conditions as well as the parameters ($r_1 = r_2 = 0.5$, $\alpha_{12} = 0$, $\alpha_{21} = 0.5$, and $K_1 = K_2 = 1000$) into equation (3.14) predicts the numbers of species in the next time step:

$$\begin{aligned}
 n_1(1) &= n_1(0) + r_1 n_1(0) \left(1 - \frac{n_1(0) + \alpha_{12} n_2(0)}{K_1} \right) \\
 &= 50 + 0.5 \times 50 \left(1 - \frac{50 + 0 \times 1200}{1000} \right) \\
 &= 73.75, \\
 n_2(1) &= n_2(0) + r_2 n_2(0) \left(1 - \frac{n_2(0) + \alpha_{21} n_1(0)}{K_2} \right) \\
 &= 1200 + 0.5 \times 1200 \left(1 - \frac{1200 + 0.5 \times 50}{1000} \right) \\
 &= 1065.
 \end{aligned}$$

Thus, the tip of the arrow is placed at $\{73.75, 1065\}$. The same procedure is followed for as many arrows as desired (100 in this plot). It should be noted that sometimes the change per time step is too short or too long to plot with an arrow in this way. Thus, the arrows are often rescaled by a factor to make their lengths more visually appealing.

Vector-field plots are a great way to visualize interactions between two variables. In this example, the arrows help us to see that the variables tend to converge to the equilibrium (arrows point toward the filled circle), suggesting

A *vector-field plot* places arrows on a phase-plane diagram to indicate the direction of change of the system.

that the equilibrium is stable. Also, the variables do not spiral around the equilibrium, so we would not expect cycling. It is also possible to see that the number of species 2 (position along the vertical axis) can rise above its equilibrium value of 500 before falling back down as the number of species 1 rises (position along the horizontal axis), as observed in Figure 4.8.

4.4.4 Null Clines

A *null cline* is a curve on a phase-plane diagram that indicates when one variable remains constant. Different variables typically have different null clines.

Another important feature that is often added to phase-plane diagrams are curves indicating when each variable remains unchanged from one time point to the next. These are known as *null clines*. Along the null cline corresponding to the variable on the horizontal axis, the system can move only up or down, without any horizontal movement. Similarly, along the null cline corresponding to the variable on the vertical axis, the system can move only left or right, without any vertical movement.

Using null clines, we can distinguish regions of the phase plane in which each variable grows or shrinks. For example, from the arrows in the vector-field plot (Figure 4.15), we can tell that the number of species 1 increases whenever its current value is low (left side of plot) and decreases whenever its current value is high (right side of plot). The null cline for species 1 will tell us exactly where the cutoff lies between these two regions. Similarly, the number of species 2 increases whenever its current value is low (bottom of plot) and decreases whenever its current value is high (top of plot), and the null cline for species 2 will tell us exactly where these two regions meet.

Let us see how to identify the null clines in the Lotka-Volterra model. From equation (3.14), the change in species 1 is

$$\Delta n_1 = n_1(t + 1) - n_1(t) = r_1 n_1(t) \left(1 - \frac{n_1(t) + \alpha_{12} n_2(t)}{K_1} \right). \quad (4.3a)$$

When Δn_1 equals zero, species 1 remains at a constant population size. There are three ways for (4.3a) to equal zero: if $r_1 = 0$, if $n_1 = 0$, or if $1 - (n_1(t) + \alpha_{12} n_2(t))/K_1 = 0$. The first condition is known as a *special case of the parameters*; it will not usually be of biological interest, unless that special case happens to pertain to a species. The second condition occurs when species 1 is absent, indicating that the vertical axis at $n_1 = 0$ is a null cline for species 1. The third condition gives us a more interesting null cline for species 1. Using the parameters $\alpha_{12} = 0$ and $K_1 = 1000$, the third condition reduces to $1 - n_1(t)/1000 = 0$, which is satisfied when $n_1(t) = 1000$ (see dashed vertical line in Figure 4.16).

Similarly, species 2 remains at a constant population size when Δn_2 equals zero, where

$$\Delta n_2 = n_2(t + 1) - n_2(t) = r_2 n_2(t) \left(1 - \frac{n_2(t) + \alpha_{21} n_1(t)}{K_2} \right). \quad (4.3b)$$

There are three ways for (4.3b) to equal zero: $r_2 = 0$, $n_2(t) = 0$, or $1 - (n_2(t) + \alpha_{21} n_1(t))/K_2 = 0$. Again, the first corresponds to a special case of the parameters,

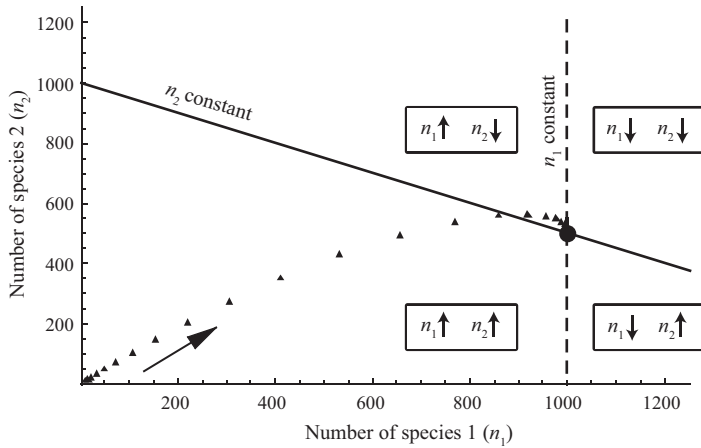


Figure 4.16: Null clines for the two-species Lotka-Volterra model of competition. The dashed line indicates when the number of species 1 remains the same in the next time step. The solid line indicates when the number of species 2 remains the same in the next time step. Only when both species remain constant over time is the system at equilibrium (filled circle). All of the parameters are the same as in Figure 4.14 with α_{12} set to 0, and the triangles again represent 50 iterations starting from $n_1(0) = n_2(0) = 10$.

and the second corresponds to extinction of species 2. The third condition is more interesting. Using the parameters $\alpha_{21} = 0.5$ and $K_2 = 1000$, the third condition tells us that the number of species 2 remains constant when $1 - (n_2(t) + 0.5 n_1(t))/1000 = 0$. We need to rearrange this equation to get an equation for $n_2(t)$ as a function of $n_1(t)$, so we can plot the null cline on Figure 4.16. Doing so, we get $n_2(t) = 1000 - 0.5 n_1(t)$, which is a line that decreases from 1000 at $n_1(t) = 0$ to 0 at $n_1(t) = 2000$ (see solid line in Figure 4.16).

Each null cline tells us when one of the two species remains constant in number. Both species remain constant in number, however, only where two null clines cross. This occurs at the equilibrium that we previously identified ($n_1 = 1000$; $n_2 = 500$). At any other point along the null cline for species 1 (dashed line), species 1 remains constant in size, but species 2 changes in number. As a result, the system gets knocked off the null cline.

It is sometimes possible to use phase-plane plots to draw more general conclusions about a model that go beyond a specific set of parameters. As one moves up in a phase-plane plot, the predicted change in a variable always has the same sign until one of its null clines is reached. Similarly, as one moves to the right along the horizontal axis, the predicted change in a variable always has the same sign until hitting one of its null clines. Only at a null cline do we see flips in the direction of change of the variable. This follows from the fact that, for a variable to change from increasing to decreasing (or vice versa), its rate of change per unit time must pass through zero. When it equals zero exactly, the variable will lie on a null cline. If we can determine the direction of change on either side of each null cline, then we can infer the behavior of the system across the entire phase plane, often for arbitrary values of the parameters.

We again consider the Lotka-Volterra model. At the null cline of species 1, $1 - (n_1(t) + \alpha_{12} n_2(t))/K_1$ equals zero. If we decrease the number of species 1 by a little bit, $1 - (n_1(t) + \alpha_{12} n_2(t))/K_1$ becomes positive, because we are now subtracting off a smaller number, $-n_1(t)/K_1$, from 1. Conversely, if we increase the number of species 1, $1 - (n_1(t) + \alpha_{12} n_2(t))/K_1$ becomes negative. The sign of this term, along with the sign of the growth rate r_1 , determines the direction of change of species 1 (see equation (4.3a)). Assuming that the growth rates are

positive, the above logic indicates that the number of species 1 will increase to the left of its null cline and decrease to the right of its null cline (dashed line). This allows us to write $n_1 \uparrow$ in the two boxes on the left of Figure 4.16 and $n_1 \downarrow$ in the two boxes on the right. The same procedure can be applied to species 2 to infer that it increases below its null cline ($n_2 \uparrow$) and decreases above its null cline ($n_2 \downarrow$). From these boxes, we can infer the direction in which the system will move. For example, in Figure 4.16, we can infer that the system will move toward the equilibrium. (Movement away from an equilibrium is also possible, depending on how the null clines cross.)

There are two caveats that must be added to the above discussion. First, even if we know the direction of change in each quadrant, the changes can be so large in magnitude in a discrete-time model that the variables overshoot their null clines. Because of this possibility, we cannot use a phase-plane analysis to determine whether a system approaches an equilibrium or oscillates around it in a discrete-time model. We will return to this issue in Chapter 8.

Second, we have implicitly assumed that the difference or differential equations are *continuous* in the variables and do not show any “discontinuities.” Discontinuities, i.e., jumps or kinks in the trajectory of a variable, are not typical in biological models but can arise when change is described by different rules for different values of the variables. For example, you could build a model with the rule that growth of species 1 is exponential at rate $+r$ when the number of species 2 is less than or equal to some threshold τ , and is exponential at rate $-r$ when there are more than τ members of species 2. This model is perfectly acceptable, but we must account for the possibility that a variable could change from increasing to decreasing without passing through zero on either side of the discontinuity at τ . By drawing discontinuities as well as null clines on a phase plane, one can infer the regions in which each variable increases or decreases over time.

4.4.5 Phase-Plane Diagrams in Continuous-Time Models

The phase-plane diagrams that we have considered so far illustrate dynamics in discrete-time models, but the methods are similar for drawing phase-plane diagrams in continuous time. Rather than iterating recursion equations, differential equations must be solved numerically, and the results used to plot changes over time in the two variables. Alternatively, we can generate a vector-field plot directly from the differential equations by drawing arrows throughout the phase plane. Each arrow begins at some point $\{n_1, n_2\}$ and ends at the point $\{n_1 + dn_1/dt, n_2 + dn_2/dt\}$ (stretched or shrunk, if necessary, to make the arrows visible). Finally, we can add null clines to a phase-plane diagram by determining the curves along which dn_1/dt and dn_2/dt are zero.

In the on-line Lab exercises on the Lotka-Volterra model of competition, you can compare the dynamics of the discrete-time and continuous-time versions. As you will see, these figures are qualitatively similar, as long as the intrinsic growth rates of both species are low.

As another example, we consider phase-plane diagrams for the continuous-time predator-prey model described by equations (3.18). In Figure 4.17, the

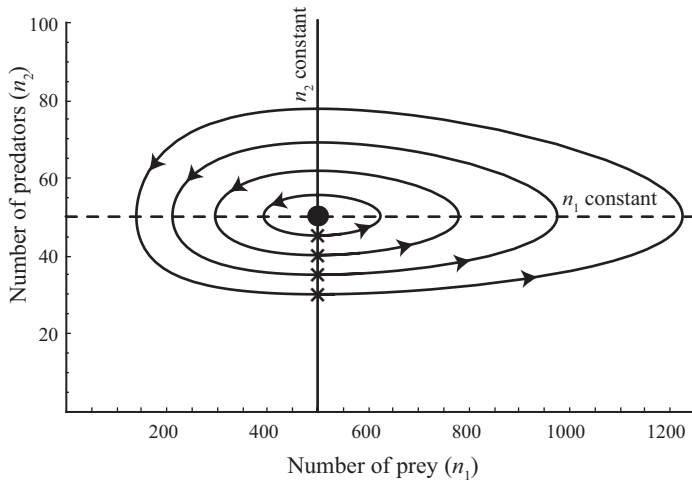


Figure 4.17: Dynamics of the predator-prey model in continuous time. The differential equations (3.18) were numerically solved using NDSolve in *Mathematica*. The resulting number of predators was plotted against the number of prey over time. The parameters used were: $r = 0.5$, $a = 1$, $c = 0.01$, $\varepsilon = 0.02$, $\delta = 0.1$, with an initial number of prey equal to $n_1(0) = 500$ and an initial number of predators $n_2(0)$ set to 30, 35, 40, or 45 ($\times$'s). The direction in which time flows is indicated by arrows. Null clines are also shown indicating where the number of prey is constant ($dn_1/dt = 0$; dashed line) or the number of predators is constant ($dn_2/dt = 0$; solid line).

horizontal axis represents the number of prey, and the vertical axis represents the number of predators. Each curve represents a numerical solution to the differential equations, using the same parameter values but starting from different initial conditions ($\times$'s). In this case, the curves form rings around a point with 500 prey and 50 predators, and the system retraces the same cycle over time. The vector-field plot suggests that this type of dynamic holds in general (at least for these parameter values) (Figure 4.18).

Figures 4.14–4.18 imply that different forms of species interactions can lead to vastly different dynamics and suggest that interactions at the same trophic level (i.e., competition) more readily generate stable equilibria than interactions across trophic levels (i.e., predation). This speculation is based, however, on only a small set of parameter values. We might see the opposite behavior with different parameter values, and it would be impossible to make graphs with every possible combination of parameters to verify this claim. To move beyond speculation, we need to learn methods to demonstrate when an equilibrium

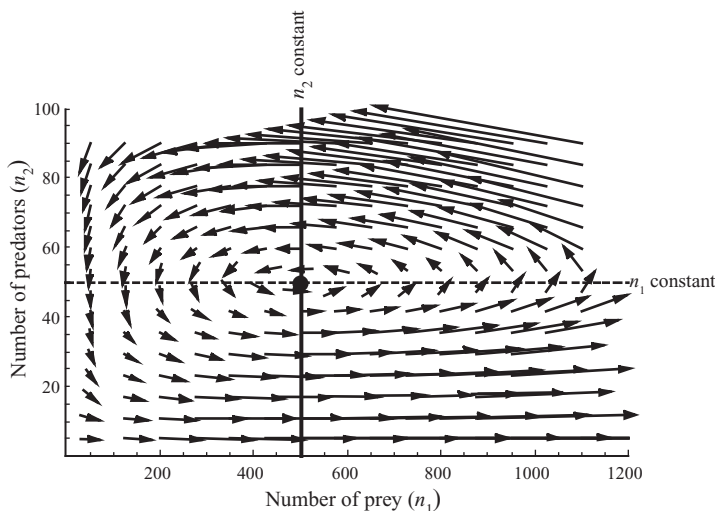


Figure 4.18: Vector-field plot for the predator-prey model in continuous time. The rate of change in the number of prey and the number of predators is shown from a series of different starting positions. Each arrow starts at a point $\{n_1, n_2\}$ and ends at $\{n_1 + dn_1/dt, n_2 + dn_2/dt\}$. All parameters and null clines are the same as in Figure 4.17.

is stable (Chapter 5) and to determine when cycling is possible (Chapters 8 and 11).

Interestingly, we can use phase-plane diagrams to understand why chaotic dynamics cannot be observed in continuous-time models with only two variables (Box 4.1). Imagine starting two replicate trajectories initially close to one another somewhere within a phase-plane diagram. These trajectories can potentially take on many different shapes, but it is not possible for the two trajectories to cross in a continuous-time model. If they were to cross, then the values of the variables would be exactly the same for both trajectories at the point in time when the trajectories cross. But from this point on, the two replicates would have identical dynamics in any deterministic model (i.e., as long as chance plays no role in the model). This fact ensures that there will always be order to trajectories started nearby one another in a two-variable continuous-time model. The trajectories might form closed orbits (as in Figure 4.17), they might spiral in together, or they might spiral out together. It is not possible, however, for two replicate trajectories to crisscross each other. Adding a third variable to a continuous-time model, however, releases the dynamics from this constraint. At an intuitive level, two trajectories can now “cross” in a phase-plane diagram of two of the variables, as long as they have different values of the third variable. Indeed, continuous-time models that exhibit chaos with more than two variables have been described in the biological literature, e.g., in food-web models involving three species (Hastings and Powell 1991; Klebanoff and Hastings 1994).

Example: Dynamics of HIV after Initial Infection (Revisited)

Now that we have a suite of useful graphical techniques for analyzing models, let us return to the model of Phillips (1996) for the dynamics of HIV within the human body. Recall that the concentration of HIV in the blood during the first 100–150 days of an infection rapidly increases, reaches a maximum, and then tapers off (Figure 1.3). Phillips’ model explored the possibility that this decline in HIV concentration is due to the depletion of uninfected lymphocytes.

We begin by numerically solving the differential equations in Box 2.4, using a mathematical software package such as *Mathematica*. Figure 1.5 presented a plot of the number of virions (virus particles) in the bloodstream over time using the parameter estimates of Table 1 in Phillips (1996). Figure 4.19a presents the same plot, along with the corresponding plots for the number of latently infected and actively infected cells (see Figure 1 of Phillips (1996)).

One of the most useful applications of the graphical techniques outlined in this chapter is to see how the model’s predictions depend on parameter values. Phillips (1996) obtained independent estimates for most of the parameter values used in the model, but there were some parameters for which little information was available. For example, the rate at which virions infect susceptible CD4⁺ cells per day, β , was unknown. Given the importance of this parameter in HIV transmission among cells, we might expect errors in the estimate of β to have a large impact on the model’s predictions.

The sensitivity of the results can be checked using the graphical techniques of section 4.2. First, suppose that the value of β used in Figure 4.19a is an order

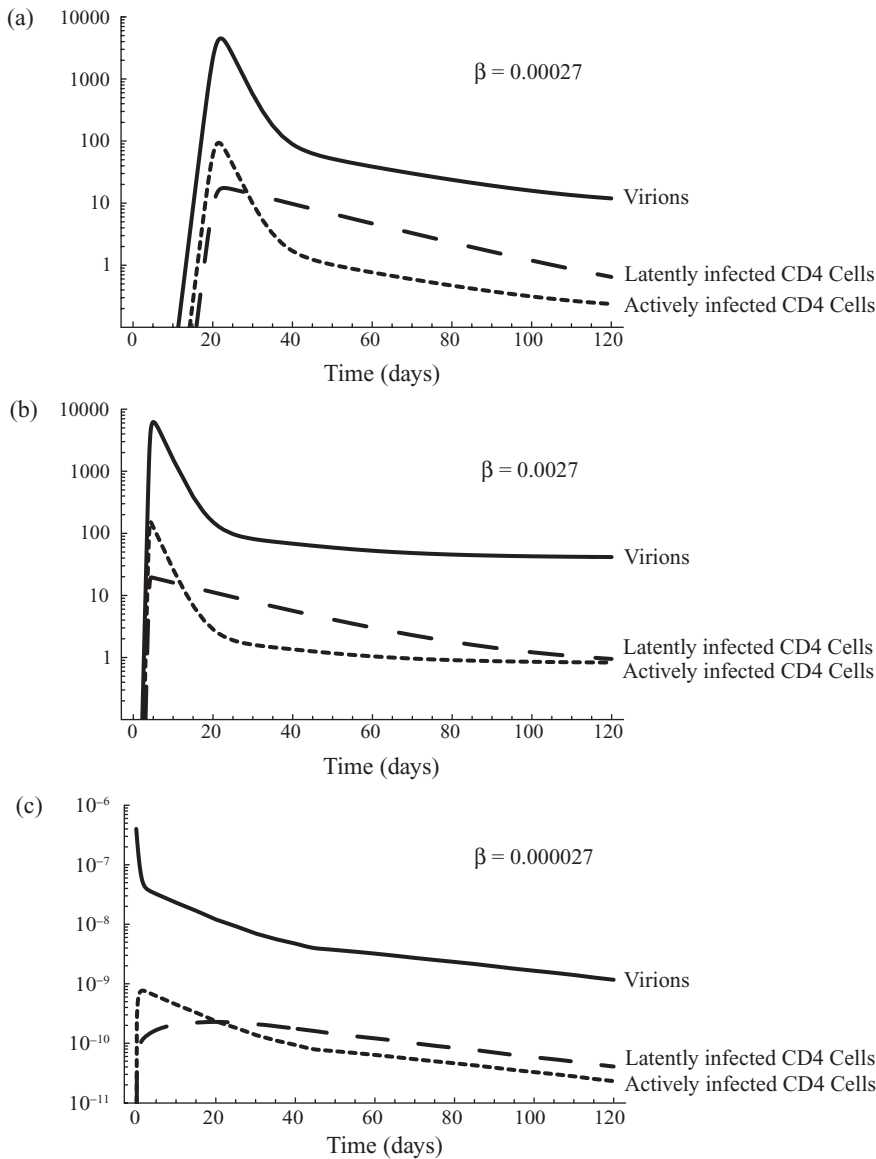


Figure 4.19: HIV dynamics within the bloodstream. The number of virions, latently infected CD4+ cells, and actively infected CD4+ cells are plotted over the first 120 days of an HIV infection. Results are based on simulation of the equations in Box 2.4 with the following parameter values from Phillips (1996): $\Gamma = 1.36$, $\mu = 1.36 \times 10^{-3}$, $\tau = 0.2$, $p = 0.1$, $\alpha = 3.6 \times 10^{-2}$, $\sigma = 2$, $\delta = 0.33$, $\pi = 100$, along with (a) $\beta = 0.00027$, (b) $\beta = 0.0027$, or (c) $\beta = 0.000027$ (note change in vertical axis). The initial number of virions was 4×10^{-7} per mm^3 of blood and the initial number of susceptible cells was 1000τ per mm^3 of blood (no infected cells were initially present).

of magnitude too low (i.e., it is 10 times too small). This amount of uncertainty is not uncommon in such parameters. If we change the value of β from $\beta = 0.00027$ to $\beta = 0.0027$, the predictions of Figure 4.19b result. The model still does a reasonable job of explaining the qualitative pattern that is observed in the virus concentration in the blood (i.e., the rise and fall). The model also

makes similar predictions regarding the maximum viral load that is reached. What has changed significantly, however, is the amount of time that it takes to reach the maximum viral load. The increased transmission coefficient β has resulted in the viral concentration reaching its peak much earlier, around 5 days post infection rather than around 21–22 days post infection.

In the same way we can determine how the predictions are altered if β is decreased by an order of magnitude, to $\beta = 0.000027$. Interestingly, doing so causes the number of virions in the blood to immediately decay to zero (Figure 4.19c). This reveals that the qualitative predictions of the model change dramatically as β is decreased below some critical value. In this case the infection is immediately eliminated from the body. This makes sense—if the virions are not effective enough at infecting susceptible CD4+ cells, then the body will be able to clear HIV out of the bloodstream. This suggests that we try to analyze the model mathematically in order to characterize when we expect the model to exhibit one behavior (establishment of the virus) versus another (elimination of the virus). In Chapter 8 we will develop mathematical techniques, known as stability analyses, for doing exactly that. For now, however, we can numerically simulate the dynamics using a range of β to find that there is a critical value of β near $\beta = 0.000034$. Above this value the virus rises and falls in concentration as is observed in a typical infection (as in Figures 4.19a and 4.19b), but below this value the virus is cleared from the body (as in Figure 4.19c).

Graphical techniques can also suggest which mathematical analyses might help answer particular questions of interest. For example, Figure 4.19 suggests that the curves for the concentration of virions in the blood and for the concentration of actively infected CD4+ cells are almost exactly proportional to one another. Put another way, the concentration of actively infected CD4+ cells is approximately some constant fraction k of the concentration of virions in the blood. Mathematically, this means that, although the concentration of virions, V , and the concentration of actively infected CD4+ cells, E , both change through time, the approximate relationship $V \approx k E$ appears to hold. Why might this be so? The reason is closely tied to an important mathematical technique known as *separation of time scales*. This technique will be developed in greater detail in Chapter 9, but let us get a flavor of it here.

To understand why the concentration of virions and actively infected CD4+ cells might always differ by a constant factor we need to consider the underlying differential equations of the model in more detail (Box 2.4). Although the four differential equations of the model allow all four variables to change simultaneously through time, some variables might change faster than others depending upon the parameter values. Examining the parameter values for the HIV dynamics (see caption to Figure 4.19), the parameters π and σ in the differential equation for the viral load (i.e., dV/dt) are much larger than the other parameters, by about two orders of magnitude. Therefore, we might expect the dynamics of the viral load to operate much faster (i.e., on a shorter time scale) than those of the other variables. This would cause the viral population size to reach an approximate steady state given the current values of the other variables. As the other variables change slowly, the viral load will then reequilibrate quickly and reach a new steady state corresponding to the new combination of

the other variables. Consequently, we might expect the viral population size to rapidly track changes in the other variables.

If the viral load reequilibrates rapidly, we would expect V to be approximately constant at any point in time. That is, we would expect $dV/dt = \pi E - \sigma V$ to be nearly zero at all times. Solving $\pi E - \sigma V = 0$ for V gives us a steady state solution for the number of viruses in the bloodstream: $V = \pi E/\sigma$. This steady state solution depends only on the number of actively infected CD4+ cells, E , explaining why these two variables change in parallel. Furthermore, this relationship accurately predicts that the number of virions should be $\pi/\sigma = 50$ times the number of actively infected CD4+ cells, which is a perfect match to the graphs (Figure 4.19). This example illustrates how patterns apparent in graphs can be used to guide us toward appropriate mathematical analyses.

4.5 Concluding Message

In this chapter we have discussed three graphical techniques that allow us to obtain a good qualitative understanding of a model's behavior without conducting any formal mathematical analyses. Plotting a variable over time illustrates the temporal dynamics of a model, giving a good sense of how a model behaves (section 4.2). Plotting a variable in the next time step (or the change in the variable) versus the current state of the variable indicates the direction in which the system tends to move (section 4.3). Such plots also allow us to identify the equilibria of a model, defined as those points where the system remains unchanged. Phase-line and phase-plane diagrams also illustrate the direction of change as a function of the current state of the system (section 4.4). When a model contains two variables, phase-plane diagrams are particularly useful, as these diagrams indicate how interactions between the variables influence the behavior of the system.

The above graphical techniques allow us to build our intuition about a model, and they provide a way of communicating this understanding to others. There is wisdom in the old adage: a picture is worth a thousand words. Nevertheless, while graphical approaches are useful, they do have the drawback of requiring us to specify parameter values, and they have a limited capacity for providing very general results. For this we must learn mathematical techniques for analyzing models.

Problems

Problem 4.1: Plot the population size $n(t)$ with respect to time using the recursion equation for logistic growth, $n(t+1) = n(t) + r n(t) (1 - n(t)/K)$, for the populations of yeast studied by Mable and Otto (2001). Use the parameters estimated from the growth of (a) haploid yeast ($K_{\text{haploid}} = 3.7 \times 10^8$, $r_{\text{haploid}} = 0.55$, $n(0) = 3.0 \times 10^5$) and (b) diploid yeast ($K_{\text{diploid}} = 2.3 \times 10^8$, $r_{\text{diploid}} = 0.55$, $n(0) = 1.9 \times 10^5$). Compare your figures to Figure 3.4.

Problem 4.2: Plot the population size $n(t)$ with respect to time using the differential equation for logistic growth, $dn/dt = r n(t) (1 - n(t)/K)$, for the populations of yeast studied by Mable and Otto (2001). Use the parameters estimated from the

growth of (a) haploid yeast ($K_{\text{haploid}} = 3.7 \times 10^8$, $r_{\text{haploid}} = 0.55$, $n(0) = 3.0 \times 10^5$) and (b) diploid yeast ($K_{\text{diploid}} = 2.3 \times 10^8$, $r_{\text{diploid}} = 0.55$, $n(0) = 1.9 \times 10^5$). Compare your figures to Figure 3.4.

Problem 4.3: Plot the population size $n(t)$ with respect to time using the recursion equation for logistic growth, $n(t + 1) = n(t) + r n(t) (1 - n(t)/K)$. Starting with a population of size 100, show that the population grows in a qualitatively similar manner regardless of the carrying capacity (use $K = 500$, $K = 1000$, and $K = 5000$) but with drastically different behavior depending on the growth rate r . That is, show that (a) the population smoothly approaches the carrying capacity without overshooting it when $r = 1.0$, (b) the population settles upon a two-point cycle when $r = 2.1$, (c) the population settles upon a four-point cycle when $r = 2.5$, and (d) the population varies erratically and is chaotic when $r = 2.7$.

Problem 4.4: Figure 4.20 illustrates the diploid recursion equation (3.13a) as a function of the allele frequency $p(t)$ for a different set of fitnesses than Figure 4.10. (a) Place circles on the diagonal line indicating where the three equilibria are. (b) Starting from an allele frequency of 0.1, use the cobwebbing procedure of Box 4.2 to predict the allele frequency after three generations of selection. (c) Draw a phase-line plot with arrows showing the direction of allele frequency change that is consistent with figure 4.20. (d) For each of the three equilibria, specify whether it is locally stable or unstable.

Problem 4.5: In a plot of the allele frequency in one generation, $p(t + 1)$, versus the allele frequency in the previous generation, $p(t)$, under haploid selection, prove that the slope is never negative. That is, use the recursion equation

$$p(t + 1) = \frac{W_A p(t)}{W_A p(t) + W_a (1 - p(t))}$$

to calculate and factor the derivative $dp(t + 1)/dp(t)$ which describes the slope in this plot. By examining the sign of each factor, show that this derivative must always be positive.

Problem 4.6: In a plot of the allele frequency in one generation, $p(t + 1)$, versus the allele frequency in the previous generation, $p(t)$, under diploid selection, prove that the slope is never negative. That is, use the recursion equation

$$p(t + 1) = \frac{W_{AA} p(t)^2 + W_{Aa} p(t)(1 - p(t))}{W_{AA} p(t)^2 + W_{Aa} 2p(t)(1 - p(t)) + W_{aa} (1 - p(t))^2}$$

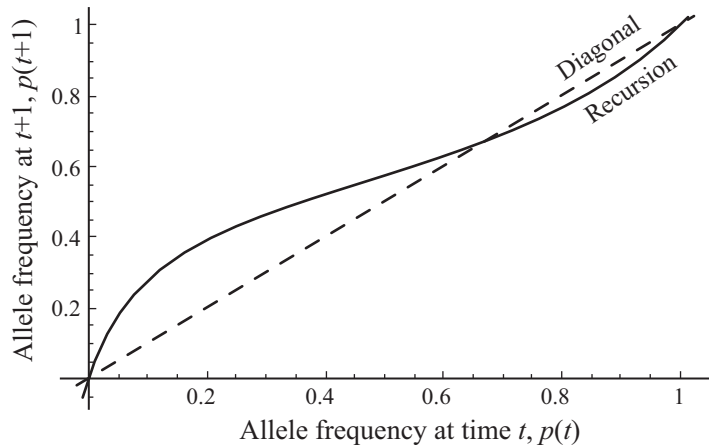


Figure 4.20: Allele frequency recursion in the diploid model of selection

to calculate and factor the derivative $dp(t+1)/dp(t)$, which describes the slope in this plot. By examining the sign of each factor, show that this derivative must always be positive.

Problem 4.7: Find the equations for the null clines of the Lotka-Volterra predator-prey model described by equation (3.18):

$$\begin{aligned}\frac{dn_1}{dt} &= r n_1 - a c n_1 n_2, \\ \frac{dn_2}{dt} &= \varepsilon a c n_1 n_2 - \delta n_2.\end{aligned}$$

Specify which equations must be satisfied for the number of prey (n_1) to remain constant and which equations must be satisfied for the number of predators (n_2) to remain constant.

Problem 4.8: In the consumer-resource model described by equations (3.17), resources (n_1) enter the system at a constant rate and are continuously eaten by a consumer species (n_2):

$$\begin{aligned}\frac{dn_1}{dt} &= \theta - a c n_1 n_2, \\ \frac{dn_2}{dt} &= \varepsilon a c n_1 n_2 - \delta n_2.\end{aligned}$$

For example this model might describe the flow of insect larvae into a pond containing minnow predators. Assume that 1000 new larvae arrive per minute ($\theta = 1000$), that the per capita contact rate is $c = 0.01$ per minute, that the probability that a minnow will consume a larva once detected is $a = 1$, that one larva is the energetic equivalent of $\varepsilon = 0.0005$ minnows, and that the per capita death rate of minnows is $\delta = 0.001$. (a) Draw a vector-field plot for this model. (b) Determine the null clines for this model and add them to the plot. (c) Describe what happens to the number of larvae and minnows over time, and discuss any limitations of the vector-field plot in this case.

Problem 4.9: The logistic equation (3.5a) differs from the recursion equation often considered in mathematical discussions of chaos, which is given by

$$x(t+1) = \mu x(t) (1 - x(t)).$$

Show that if we define $x(t) = (\mu - 1) n(t)/(\mu K)$ and $\mu = 1 + r$, we can rearrange the above equation to get (3.5a). This “transformation” explains why the transition to chaos occurs at $r^* = 2.569944$ in this book but at $\mu^* = 3.569944$ in other treatments.

Problem 4.10: Use a mathematical software package to plot the differential equations for the Lotka-Volterra model of competition between two species given by (3.15):

$$\begin{aligned}\frac{dn_1}{dt} &= r_1 n_1(t) \left(1 - \frac{n_1(t) + \alpha_{12} n_2(t)}{K_1} \right), \\ \frac{dn_2}{dt} &= r_2 n_2(t) \left(1 - \frac{n_2(t) + \alpha_{21} n_1(t)}{K_2} \right).\end{aligned}$$

Assume that $r_1 = r_2 = 0.5$, $\alpha_{12} = 0$, $\alpha_{21} = 0.5$, and $K_1 = K_2 = 1000$ and that the initial population size ten for each species ($n_1(0) = 10$, $n_2(0) = 10$). Compare the

resulting graphs to Figure 4.8. Do the discrete-time and continuous-time graphs differ substantially for this example?

Problem 4.11: After recovering from a cold, we are susceptible to catching another cold, because many different viruses can cause colds. For such diseases, the Susceptible-Infected-Recovered (SIR) model can be simplified to a Susceptible-Infected (SI) model, where individuals that recover from a cold return to the susceptible class. We can modify the differential equations for the SIR model (3.19) to describe the number of individuals susceptible to colds, S , and infected with colds, I :

$$\frac{dS}{dt} = \theta - dS(t) - acS(t)I(t) + \rho I(t),$$

$$\frac{dI}{dt} = acS(t)I(t) - \delta I(t) - \rho I(t).$$

Let $\theta = 4$ be the rate at which individuals enter the population, $c = 0.01$ be the per capita rate of contact between a susceptible and an infected individual, $a = 0.1$ be the probability of transmission of the disease per contact, $\rho = 0.2$ be the rate at which infected individuals recover from a cold, and $d = 0.01$ and $\delta = 0.02$ be the death (or emigration) rates of susceptible and infected individuals. Figure 4.21 illustrates a vector-field plot for this model.

Each arrow starts at an (x, y) coordinate equal to some combination of (S, I) and points in the direction of $(S + dS/dt, I + dI/dt)$. To make the arrows easier to see, each arrow was drawn with a constant length of five.

(a) Determine the null-cline along which the number of susceptible individuals, S , remains constant. Write this null cline in terms of what its height must equal ($I = \dots$). (b) Determine the null cline along which the number of infected individuals, I , remains constant. Write this null cline in terms of what its height must equal ($I = \dots$). (c) Add curves for these null clines to the figure, specifying what remains

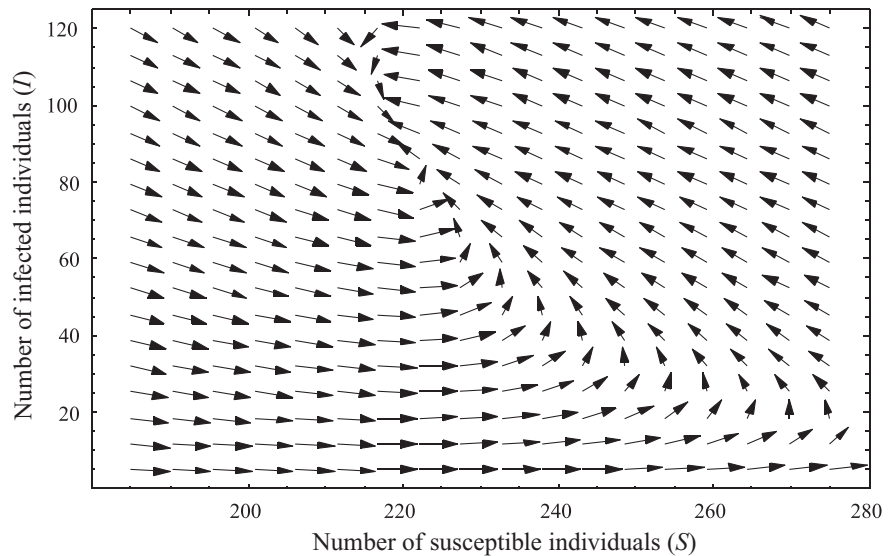


Figure 4.21: Vector-field plot for the susceptible-infected model

constant along each curve. (d) Add a dot to the figure where both variables remain constant (the equilibrium). Convince yourself that if the null clines have been drawn correctly, any arrow starting on a null cline where S is constant should point straight up or down and any arrow starting on a null cline where I is constant should point either left or right.

Problem 4.12: When simulating the logistic model

$$n(t + 1) = n(t) + rn(t)\left(1 - \frac{n(t)}{K}\right)$$

we found that extinction ensues for values of r greater than 3 (Figure 4.2d) because the population can overshoot the carrying capacity to such a large degree that the population size declines below zero in the next generation. (a) Find the population size $n(t)$ that leads to the maximum population size in the next generation, $n(t + 1)$. [Hint: Use the derivative of $n(t + 1)$ with respect to $n(t)$ to find where the maximum occurs.] (b) Find the maximum population size that can be reached by determining $n(t + 1)$ starting from the $n(t)$ found in (a). (c) Find $n(t + 2)$ starting from the maximum population size $n(t + 1)$. (d) Factor the result from (c) and show that $n(t + 2)$ will be less than zero and the population will go extinct if r is greater than 3.

Problem 4.13: As suggested by Figure 4.19, we can approximate Phillips' HIV model by assuming that the number of virions equilibrates rapidly ($dV/dt = 0$), while the other variables change more slowly over time. This assumption allows us to reduce the number of differential equations in the model from four to three, giving

$$\begin{aligned}\frac{dR}{dt} &= \Gamma\tau - \mu R - \beta R \frac{\pi}{\sigma} E, \\ \frac{dL}{dt} &= p\beta R \frac{\pi}{\sigma} E - \mu L - \alpha L, \\ \frac{dE}{dt} &= (1 - p)\beta R \frac{\pi}{\sigma} E + \alpha L - \delta E.\end{aligned}$$

(Here, we have used the fact that dV/dt will be 0 if $V = \pi E/\sigma$ to replace V in the remaining three equations of Box 2.4.) Plot the dynamics of these differential equations over time, using the same parameters as in Figure 4.19. Describe any differences between your figure and Figure 4.19 (they should be similar).

Problem 4.14: Here, we ask whether the inclusion of the latent class of CD4+ cells is critical to the results of Phillips' model of HIV. Suppose that there is no latent class of CD4+ cells (set $L = 0$ and $p = 0$). This reduces the three-variable version of Phillips' HIV model described in Problem 4.13 to a model with two equations:

$$\begin{aligned}\frac{dR}{dt} &= \Gamma\tau - \mu R - \beta R \frac{\pi}{\sigma} E, \\ \frac{dE}{dt} &= \beta R \frac{\pi}{\sigma} E - \delta E.\end{aligned}$$

Simulate this two-variable model using the same parameter values as in Figure 4.19. Based on your results, is the latent class of cells crucial for the conclusions drawn by Phillips about the initial dynamics of the virion population?

Problem 4.15: (a) Generate a vector-field plot for the HIV model described by the two equations in Problem 4.14. (b) Find the null clines of this model and draw them on the vector-field plot.

Further Reading

For more information on chaos, consult

- Stewart, I. 1997. *Does God Play Dice? The New Mathematics of chaos*. Penguin Books, London.
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